The Semantic Ontology of Actualization

Clocks, weights, and the common architecture of relativity, thermodynamics, and quantum theory

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Abstract

Relativity, thermodynamics, and quantum theory are usually merged, when they are merged at all, by unifying their *equations*. This paper merges their *semantics* instead. We take as primitive a distinction that all three already presuppose but none formalizes: between what is actual — a finite record of events that has happened — and what is potential — a structured field of weighted alternatives that has not. An experiment is then a three-fold selection: of an intervention, of a question, and of one outcome from the weighted alternatives that question admits. The FTD-v2 successor reported here adopts a noncommutative local net for potentiality while retaining a commutative algebra of actual records, and supplies an imposed deterministic, globally contextual reference selector Σ. This closes logical compatibility, not the physical Born problem: the selector consumes state–effect weights and therefore cannot explain their substrate frequencies. The finite qutrit tensor net is only a selected Type-I consistency witness; no Type-III classification follows from it or from finite spectral statistics. Relativity constrains Σ's operational form, and we separate — as is not always done — the order-independence of spacelike instruments, which is standard physics, from strong selector-history equivariance, which is a distinct condition not imposed by v2 and is not required merely to hide a preferred foliation operationally. Thermodynamics supplies a language for irreversible records, their cost, and their timescales, while a complete entropy balance for the selector remains open. Global update order, local clock duration, and the count of compliant actualization gates are kept distinct. In particular, G^* is the exact period factor and selected gate cadence of a maintained critical quartic clock; it is neither global time nor an outcome selector. Two contributions are developed here. First, a *reorganization of the standard power-law clock quadrature*: within the homogeneous family $V = \lambda|x|^n$, the period constant factors into a quadrature root $\sqrt{\pi}$ and a degree ratio, with the harmonic case uniquely self-reproducing — which is why its constant is a power of π — and the quartic case yielding the lemniscatic $G^* = \Gamma(\frac{1}{4})/\Gamma(\frac{3}{4})$. A proposition with named hypotheses states when a threshold forces the quartic, in the $A \to 0$ limit, with its finite-amplitude failure mode exhibited exactly. In a discrete model, timekeeping then splits by a decidable rigidity criterion into three exhaustive cases; the screened native candidates realized only the harmonic one and no native quartic clock was found. A quartic clock is available in a purchased construction at an itemized price — four bodies and one additional interaction range, plus a threshold that must be actively held. Its leading $A \to 0$ normal-form period law is exact. Finite-amplitude drift has been computed only in a harmonic-bond surrogate,

*Draft, 2026-08-09. This is a foundations paper: an architecture assembled from established external results plus explicitly booked v2 axioms, adoptions, and selections, together with exactly stated and author-attested, protocol-locked measurements made across explicitly declared mechanism-level and engine-scoped instruments/model classes associated with one concrete discrete programme. It should be cited as foundations, never as a physics result. Every quantitative claim carries its scope in the text and again in the status ledger of Section 13. Every data-bearing external figure is generated by an accompanying script; conceptual diagrams are encoded directly in this source. The detector measurements of Section 8 are transcribed from the protocol records listed in Appendix A; those records have matching runner hashes but no immutable pre-run timestamp. The finite-amplitude clock surrogate and native mechanical screens of Section 4, and the quantitative results of Section 9, are exploratory computations, not pre-registered, and are marked as such where they appear.

not in the declared compact carrier. Second, a classical detector result: in a protocol-locked mechanism-level measurement, threshold statistics interpolate from amplitude weighting toward action/occupation weighting as mode frequency exceeds the noise bandwidth. We call the normalized interpolation coordinate the *occupation fraction*; it is not a probability and the experiment does not derive the Born rule. The physical Born pushforward remains open. We report with equal prominence the accompanying negative — that the same law did not transfer to the model's own thermal field in the sampled domain — and a measured, instrument-scoped purity diagnostic: in the tested additive model the slow channel must be essentially the sole stochastic element at the threshold. We claim a shared architecture with itemized interfaces, not a unification.

1 Building a world, and paying for it

A physicist moving between relativity, thermodynamics, and quantum theory changes vocabulary three times. In relativity the primitive is an *event* and the structure is a partial order. In thermodynamics the primitives are *macrostates* and the structure is a monotone. In quantum theory the primitive is a *state* and the structure is an algebra of operators. The three are routinely combined at the level of equations — quantum field theory on curved backgrounds, black-hole thermodynamics, relativistic statistical mechanics — and yet the combinations remain notoriously partial, and the partiality always localizes to the same place: what happens when something *becomes the case*.

This paper proposes that the three theories already share a semantic distinction which none of them makes explicit, and that making it explicit is what the merge requires. The distinction is between the *actual* — a finite, singular record of what has occurred — and the *potential* — a structured field of weighted alternatives that have not occurred and mostly never will. Once the distinction is drawn, each theory turns out to govern a different part of it. Quantum theory, including its relativistic form, is the law of the potential layer: how weighted alternatives are structured, how they evolve, how they compose. Relativity constrains how the transition from potential to actual can be described without a preferred frame. Thermodynamics supplies the language of direction, retention and cost. Across the concrete model and declared auxiliary instruments below we establish an operationally noninvertible update and a timescale-dependent classical detector regime; neither result by itself establishes entropy production or quantum outcome statistics.

The way to test a claim like that is to try to build the thing and keep the receipts. So the paper proceeds as a construction: begin with the distinction and nothing else, and at each stage add only what the stage before it made necessary — saying, every time, what the addition cost. A section that adds nothing is a section that should not be there, and a section that adds something without naming a price is the failure mode this method exists to prevent.

Four kinds of cost recur, and the paper uses these four words for them throughout. A **postulate** is declared outright: it buys structure and is answerable only to consistency and to the reader's judgement that it was worth declaring. An **import** is an external theorem, carried in whole and conditional on its own premises, which the paper does not re-prove. A **purchase** is a piece of machinery the substrate did not supply and that has to be added to it — an interaction range, a species, a degree of freedom. And a **maintenance** is a condition that is not paid once but held: a configuration the world drifts away from unless something keeps returning it. The last is the least familiar and turns out to be the most expensive.

The successor's axioms, adoptions, and selections are declared in Sections 2, 4, and 7. They include the potentiality net, the preparation map, split locality, measurement independence, and the reference selector; none is relabelled as emergence from the classical ternary substrate. The imports are itemized rather than summarized, because they are the largest single thing the paper does not derive. The purchases and the maintenance are what the constructive sections are about. Section 13 totals all of it, and where that table and the body disagree, the table is right.

What this paper claims, and what it does not. It claims a *common semantic architecture with itemized interfaces* — not a unification, not a derivation of quantum mechanics, and not a new dynamical theory. The reconstruction results of Section 7 are *imported* and conditional on their own premises. The detector measurements of Section 8 are made across explicitly declared mechanism-level and engine-scoped instruments/model classes associated with the concrete discrete programme, and carry the scope of their author-attested protocol records; matching hashes do not supply an immutable pre-run timestamp. The finite-amplitude clock surrogate and native mechanical screens of Section 4 are exploratory computations, not protocol-locked measurements. Planck's constant, the Hamiltonian, particle masses, and coupling constants are not derived anywhere in this paper. Nor is the lemniscatic constant of Section 4: it is the period of a shape the substrate is not shown to adopt, and it enters this framework by choice. Nor does the paper infer Born or Bell statistics from singularity, dissipation, thresholding, or noninvertibility: those properties concern how a record is made and retained, while the physical pushforward measure remains open. The full accounting is Table 2.

Successor status (FTD-0825). FTD v1 and its declined noncommutative measurement map remain historical provenance. The v2 branch instead adopts: commutative classical actual beables; a complex, noncommutative potentiality net; Moore-local substrate propagation with a globally nonseparable selector; and measurement independence $\mu(\lambda \mid C) = \mu(\lambda)$. Operational signalling remains forbidden. Reference-model success is not substrate evidence: physical Born recovery, preferred-order hiding, and native maintenance of the G^* clock remain open.

Nothing has been declared yet, and nothing bought. The construction begins in the next section with the cheapest possible commitment: a single distinction, without which there is not yet anything for a physics to be about.

2 Primitive ontology

At stage n the **actual history** is a finite sequence of events

$$H_n = (e_1, e_2, \dots, e_n). \tag{1}$$

There may be many possible continuations, collected in a set $\mathcal{P}(H_n) = \{e'_1, e'_2, \dots\}$. The founding move of this architecture is to refuse to identify these two objects:

$$\text{actual history } H_n \;\neq\; \text{potential continuations } \mathcal{P}(H_n). \tag{2}$$

Postulate 1 (Singular actuality)**.** One realized experiment produces one next event e_{n+1}, so that $H_{n+1} = H_n \circ e_{n+1}$.

Postulate 1 is a declared commitment of this scaffold, not a theorem and not a consequence of any dynamics presented here. Its principal alternative — denying it, and taking all weighted continuations to be actual — is a live position in the literature and is priced symmetrically in Section 9.2. We adopt it because the architecture we are building is an architecture *of* actualization, and a theory in which nothing becomes the case has no actualization to organize.

This commitment should not be confused with a claim that ordinary uncertainty is quantum. A bag of marbles may have a completely definite microconfiguration even when its holder does not know which marble will be drawn. The draw makes one outcome actual for the record; the other draws are counterfactual opportunities, not additional recorded events. That is a model of *uncertain discrete states*, and it is compatible with an ordinary classical probability space. Postulate 1 asserts the singularity of the record only. It neither asserts a

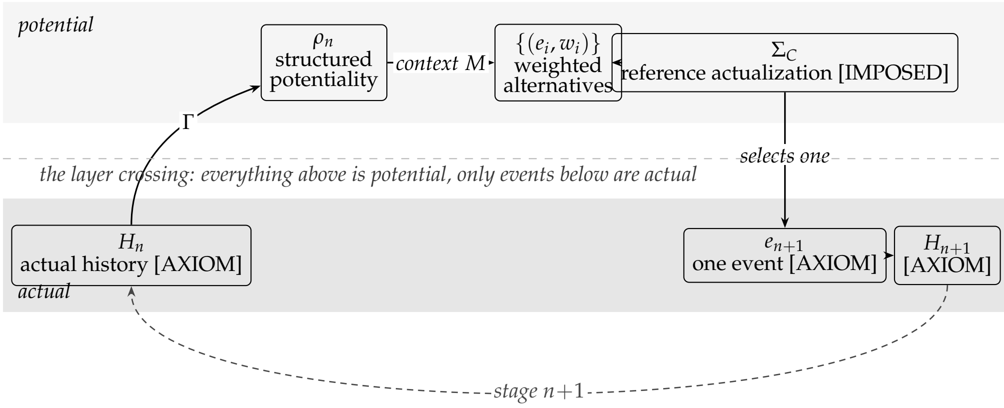

Figure 1: The shape of the recursion, drawn here before most of it has been paid for. Structure is encoded in shape and rule, never in colour: solid boxes are theorem-grade within their stated premises, while thin boxes are axioms, selections, or imported representations. Actual history generates structured potentiality through a state-assignment map Γ; a chosen context resolves it into weighted alternatives; the actualization block selects one; the record extends and the cycle repeats. Everything above the lower band is potential; only the events in it are actual. *Of the objects shown, this section has bought only the two bands and the singular event.* The state-assignment map, the context and the weights are purchased in Section 7, which is where the diagram closes into the equation it depicts. The imposed Σ_C closes only the reference recursion; its physical substrate pushforward remains open.

coherent superposition of marbles nor derives interference, contextuality, the Born rule, or a Bell violation.

An experiment is not merely an encounter with a probability distribution. Before any distribution is defined, two selections are made: of a controlled input $u = \Sigma_U(\mathcal{U})$ and of a measurement context $M = \Sigma_M(\mathcal{M})$. These are *operational* selections — a knob, a clock, a deterministic circuit will do — and carry no commitment about freedom. The point is logical order: *what are we doing to the system*, and *what distinction will be read out*, both precede *with what frequencies*.

One thing this buys is worth isolating before going on, because it is smaller than it looks. A record that extends has an order. What it does not have is a scale on that order, and the next section shows why that gap cannot be closed by looking at the record harder.

3 Succession is not yet time

What has been bought so far is a distinction and a rule for extending a record. That is strictly less than a world, and the gap is worth naming precisely, because it is the first place the construction has to spend.

Writing $H_{n+1} = H_n \circ e_{n+1}$ gives an *order*. An order supports two predicates, *before* and *after*, and it supports nothing else. In particular it does not support *how much*. Nothing in a growing list distinguishes a long interval from a short one, because the list has no second interval to compare against: each event is simply the next.

A duration is a ratio. To say that one interval is 3.7 of another is to have counted something that happened 3.7 times, so a unit is required before a duration is meaningful, and a unit requires a *recurrence* — a feature that comes back, so that returns can be counted. Monotone succession never returns. Whatever supplies the recurrence is therefore not the index; it is a physical thing inside the history, and it must be paid for.

Two consequences follow, and they set up everything after this section.

First, the demand has a definite shape: a bounded degree of freedom, a law that restores it, and a constant relating how far it swings to how long it takes. That constant is where the price will show up, because the restoring law is not free — a substrate either supplies one or does not.

Second, and less obviously, the same argument applies to the paper's own bookkeeping. The recursion this architecture will run on is indexed by the ontic integer update order n. That index is a causal ordering variable, not an elapsed duration. Operational duration requires a maintained local clock with phase $\phi_x(n)$ and inferred duration τ_x; actualization introduces the still different count k_x of phase-compliant gates. A clock is therefore not a component of the architecture among others; it is a precondition for turning update order into measured duration. That thread is picked up in Section 10, once there is a recursion on the page to be indexed.

So: what is the cheapest recurrence a discrete substrate can host, what constant does it keep, and what does it cost?

4 Time: the clock

Four questions, in the order Section 3 leaves them. What does a period cost, in the sense of what fixes its constant? What shapes can a substrate make available? Which of them, if any, is forced? And what does the shortfall between those two answers cost, and buy? Each subsection below answers exactly one of them.

4.1 The setting

The model is a discrete one: point bodies on a lattice, interacting through a compact central pair potential with a single minimum at separation r_0, and carrying a polarity that makes the interaction graph bipartite. Two features matter. Every bond in an equilibrium sits *at* its potential minimum, so the configuration carries zero actual tension; and the potential has compact support, so "bonded" is a sharp predicate. What follows is exact within that model.

4.2 What a period costs

Before asking which oscillator a substrate supplies, it is worth asking what the homogeneous power-law family charges. Take a symmetric confining potential $V = \lambda|x|^n$ and release a body of mass m from rest at amplitude A, so $E = \lambda A^n$. Energy conservation fixes the speed, $v = \sqrt{2(E-V)/m}$, and a quarter period is the time to cross from the centre to the turning point:

$$\frac{T}{4} = \int_0^A \frac{\mathrm{d}x}{v} = A\sqrt{\frac{m}{2E}}\, I(n), \qquad I(n) = \int_0^1 \frac{\mathrm{d}u}{\sqrt{1-u^n}} = \frac{1}{n}B\big(\tfrac{1}{n},\tfrac{1}{2}\big)\,. \tag{3}$$

Within this homogeneous family, everything specific to the clock is in the $I(n)$ of (3), and it factors into a square-root quadrature term and a degree-dependent term:

$$I(n) \;=\; \underbrace{\sqrt{\pi}}_{\text{quadrature root}} \cdot \underbrace{\frac{\Gamma(1/n)}{n\,\Gamma(1/n+1/2)}}_{\text{degree ratio}}\,. \tag{4}$$

The first factor is $\Gamma(\frac{1}{2})$, arising when the homogeneous integral is reduced to a Beta function. The second factor is the part that moves with the degree n. A generic mixed or nonhomogeneous potential has no single degree and need not admit this factorization; $\sqrt{\pi}$ is not claimed here as a universal charge of every mechanical clock.

n	$I(n)$	degree ratio	
2	1.5707963268	0.8862269255	$= \sqrt{\pi}/2$
3	1.4021821053	0.7910965381	
4	1.3110287771	0.7396687798	$= G^*/4$
5	1.2537306248	0.7073417591	
6	1.2143253239	0.6851096988	
8	1.1635925712	0.6564868082	

Two rows are special, and their relation is the point.

At $n = 2$ the degree ratio is $\Gamma(\frac{1}{2})/\Gamma(1) = \sqrt{\pi}/2$ — *the quadrature root again*. The harmonic case is the unique self-reproducing one, so its constant is $I(2) = \sqrt{\pi} \cdot \sqrt{\pi}/2 = \pi/2$: a power of π and nothing else. Within this family, the circle constant in the harmonic row is therefore a Beta-function identity rather than a separate geometric input.

At $n = 4$ the degree ratio is $G^*/4$, giving $I(4) = \sqrt{\pi}\, G^*/4 = \varpi/2$, half the lemniscate constant. So π and G^* are not rival constants competing to be fundamental. They are the same quadrature factor multiplied by two different degree ratios, and $G^* = 2\varpi/\sqrt{\pi}$ is that statement rearranged.

4.3 A decidable trichotomy

Let R be the rigidity matrix of a framework, whose rows are the stretch rates $\hat{u}_e \cdot (q_i - q_j)$ of each bond under a displacement field q. Expanding a path $p(d) = p_0 + d\,q + d^2 w$, the stretch of bond e is $d\,(Rq)_e + d^2[\kappa_e(q) + (Rw)_e] + O(d^3)$ with $\kappa_e(q) = |q_i - q_j|^2_\perp/2\ell_e$. Three cases are exhaustive.

Criterion 1 (The rigidity trichotomy)**.** *At zero tension, along a direction q:*

1. *if $Rq \neq 0$, the direction is first-order rigid and the energy grows as d^2: $n = 2$, a harmonic clock;*
2. *if $Rq = 0$ and $\kappa(q) \perp \operatorname{coker} R$, the flex extends through second order; it may be obstructed at higher order or extend to a finite mechanism, so the quartic term vanishes and $n \geq 6$ or $n = \infty$;*
3. *if $Rq = 0$ but $\kappa(q)$ is not orthogonal to* coker R, *the flex is* blocked, *and minimizing over w gives exactly*

$$E(d) = \lambda(q)\, d^4 + O(d^5), \qquad \lambda(q) = \frac{1}{2} \min_w \sum_e k_e \left[\kappa_e(q) + (Rw)_e\right]^2 > 0, \tag{5}$$

so $n = 4$. If $\dim \operatorname{coker} R = 1$, with generator $\hat{\omega}$, this reduces to $\lambda = \langle \hat{\omega}, \kappa \rangle^2 / (2 \sum_e \hat{\omega}_e^2 / k_e)$; the minimum-viable carrier below has precisely this one-stress form.

Hence $n = 4$ at zero tension exists if and only if *a self-stress exists and blocks the flex.*

Two corollaries follow at once: no self-stress means no $n = 4$, and actual tension means $n = 2$, since a nonzero equilibrium stress puts a d^2 term on any flex it blocks. The quartic case lives only at the degenerate point — zero actual tension with nonzero *possible* stress. This makes what is otherwise a question about potential shapes decidable by linear algebra.

The trichotomy matters because the three cases keep time differently (Figure 2). The harmonic case is isochronous and its quadrature contains π, which (4) has already explained. The quartic case is the anti-pendulum: its frequency grows with amplitude, and its period law contains the lemniscatic ratio

$$G^* = \frac{\Gamma(\frac{1}{4})}{\Gamma(\frac{3}{4})} = 2.958675119188639\ldots \tag{6}$$

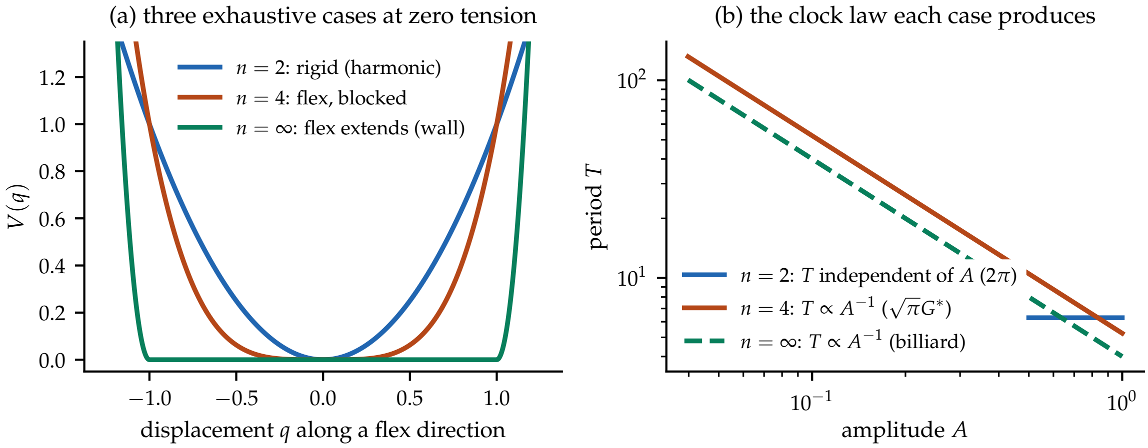

Figure 2: The rigidity trichotomy and the clocks it produces. (a) The three exhaustive cases of Criterion 1 at zero tension. (b) The corresponding period laws for the plotted normalization $m = 1$, $V = |x|^n/2$: the harmonic coefficient is 2π; the quartic coefficient is $\sqrt{\pi}\,G^*$ in $T \propto A^{-1}$; a hard wall also gives $T \propto A^{-1}$ but with no special coefficient.

4.4 What forces the quartic, and what does not

The trichotomy says a quartic clock is *available*. A stronger statement is available too, and because it is the one most likely to be over-read, it is stated with its hypotheses attached and its failure modes named.[1]

Proposition 1 (Threshold forcing). *Let a single degree of freedom evolve in a potential $V(x;\mu)$ such that*

(H1) *V is smooth in x and in a control parameter μ, with one degree of freedom;*

(H2) *V is invariant under $x \mapsto -x$;*

(H3) *the family crosses $\mu = 0$ transversally, at which point $\partial_x^2 V(0;0) = 0$;*

(H4) *the quartic coefficient is nonzero, $\lambda \equiv \frac{1}{4!}\partial_x^4 V(0;0) > 0$.*

Then at $\mu = 0$ the leading behaviour is $V = \lambda x^4 + O(x^6)$, and in the limit $A \to 0$

$$T \cdot A \;=\; \sqrt{\pi}\,G^*\sqrt{m/2\lambda}.$$

The content is in what each hypothesis excludes. (H2) is what kills the cubic — and it kills it *by symmetry, not by tuning*, which is why the quartic is generic here rather than fine-tuned. (H3) is the threshold. (H4) is nondegeneracy. And the $A \to 0$ clause is not decoration: it is the whole of the second failure mode below.

The substrate is not known to sit at a threshold. Proposition 1 forces the quartic *given* a threshold; it says nothing about why a substrate would be at one. Since $\mu = 0$ is a codimension-one surface, the generic configuration is not on it, and staying on it is a condition that must be actively held. The price is therefore a maintenance cost paid continuously, not a construction cost paid once — which is the sense in which this is a clock at the *edge* of stability rather than a clock in a well.

[1]The reading of the quartic as the clock of marginal stability, together with the period law and the anti-pendulum framing, originates in a companion manuscript on the edge-of-stability clock; what is added here is the statement as a proposition with named hypotheses, the factorisation (4), and the finite-amplitude term.

Two ways it fails, and only one of them is obvious. Dropping (H4) entirely gives a pure sextic and a single different constant, $4I(6)/\sqrt{\pi} = 2.7404387952$. That failure is visible.

The other is not, and it is a genuine counterexample to any finite-amplitude version of the proposition. Take $V = \lambda x^4 + \nu x^6$ with both terms present, and write $r = \nu A^2/\lambda$. Then

$$T \cdot A = 4\sqrt{\frac{m}{2\lambda}} \int_0^1 \frac{\mathrm{d}u}{\sqrt{(1-u^4) + r\,(1-u^6)}}, \tag{7}$$

and the constant sweeps *continuously* with r: the bracket runs 5.2441, 5.2133, 4.9589, 3.5563 at $r = 0$, 0.01, 0.1, 1. Every member of this family has $\lambda \neq 0$, so (H4) does not exclude it. What excludes it is the limit: $r \propto A^2$, so the drift vanishes with the amplitude and G^* is the $A \to 0$ endpoint, not a finite-amplitude value.

It is tempting to read this paper's own measurement as the same term, and an earlier draft did. That reading was wrong twice over, and the correction is worth more than the coincidence was. The exploratory twelve-degree-of-freedom *harmonic-bond surrogate* of Section 4.7 holds $T \cdot A$ constant only to 0.33% across a sixfold amplitude range, recovering G^* to 2×10^{-6} only in the $A \to 0$ extrapolation. But that 0.33% is an *excess*, not a deficit: $T \cdot A$ *rises* with amplitude and every recovered G^* sits *above* G^*, which in the coordinates of (7) is $r < 0$, not $r > 0$. And the excess is not mostly the sextic term at all — Section 4.7 decomposes it, and 22.7 parts in 23 are *kinetic*, a correction this family does not contain. The drift and the counterexample family are **not** one term seen twice. They are two different terms of comparable size, and only the smaller one belongs to this section.

4.5 Off the threshold, exactly

The proposition holds *at* $\mu = 0$. Since the whole price is that $\mu = 0$ must be held, what happens as it slips is not a footnote — it is the thing being paid for, and it is exactly solvable.

For the normal form $V = \frac{1}{2}\mu x^2 + \lambda x^4$ the quadrature closes in elliptic functions: the orbit is $x(t) = A\,\mathrm{cn}(\Omega t; \kappa)$ with $\Omega^2 = (\mu + 4\lambda A^2)/m$, and writing everything in terms of the modulus

$$\kappa^2 = \frac{2\lambda A^2}{\mu + 4\lambda A^2} \tag{8}$$

eliminates μ and A separately, leaving a single universal curve:

$$\boxed{T\,A\sqrt{\tfrac{2\lambda}{m}} \;=\; 4\,\kappa\,K(\kappa)} \tag{9}$$

with K the complete elliptic integral of the first kind. Every origin-crossing oscillation (and, for $\mu < 0$, every over-barrier oscillation) lies on this one curve: the harmonic side at $\kappa^2 \to 0$, the double-well side at $\frac{1}{2} < \kappa^2 < 1$, and the quartic clock at exactly $\kappa^2 = \frac{1}{2}$, where $4\kappa K = \sqrt{\pi}\,G^*$ and (10) is recovered.

Two things follow, and the second is the more interesting.

The threshold is a self-dual point. $\kappa = 1/\sqrt{2}$ is the fixed point of the exchange $\kappa \leftrightarrow \kappa'$ between a modulus and its complement, where $K' = K$. So G^* is not merely the constant of one potential among many; it is the value the period takes at the one modulus that is its own complement. This is a distinct self-duality: the quartic threshold lands on the complementary-modulus fixed point $\kappa^2 = 1/2$, whereas Section 4.2's harmonic self-reproduction is the recurrence of $\Gamma(1/2) = \sqrt{\pi}$. Both are fixed-point structures, but they are not the same identity.

The constant drifts two ways, and both are now exact. Detuning moves κ^2 off $\frac{1}{2}$ along (9); a sextic admixture moves the constant along (7) in r. They are independent perturbations — one changes the quadratic coefficient, the other adds a higher one — and G^* is the corner $s \to 0$, $r \to 0$ of the reduced plane, where $s = \mu/2\lambda A^2$ and $r = \nu A^2/\lambda$.

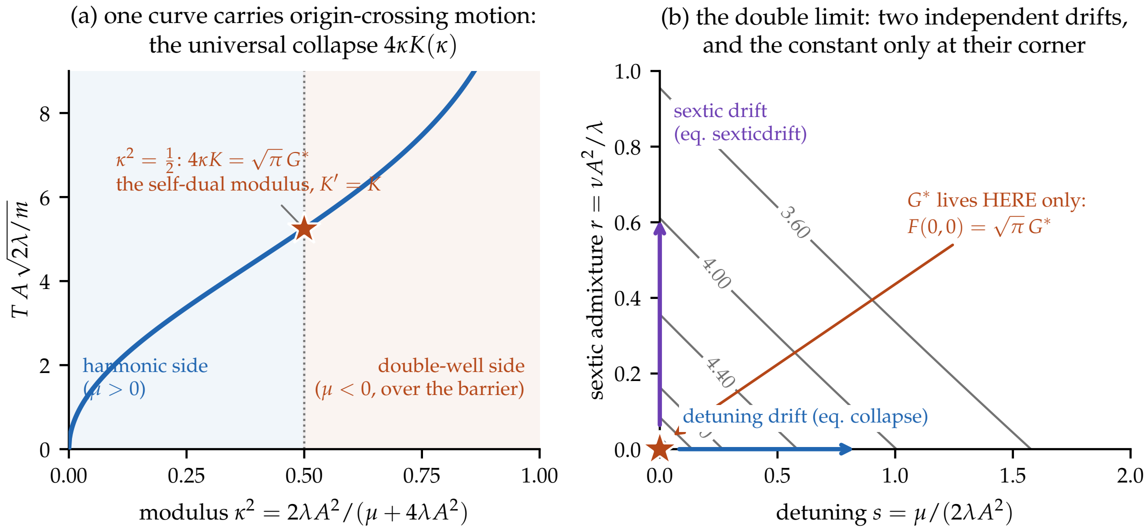

Figure 3: The crossover, exactly, and the double limit. (a) Every origin-crossing oscillation of the normal form (including over-barrier motion for $\mu < 0$) lies on the single curve $TA\sqrt{2\lambda/m} = 4\kappa K(\kappa)$: harmonic side as $\kappa^2 \to 0$, oscillation over the double-well barrier for $\kappa^2 > \frac{1}{2}$, and the quartic clock at exactly $\kappa^2 = \frac{1}{2}$ — the self-dual modulus $K' = K$ — where the value is $\sqrt{\pi}\,G^* = 5.2441$. (b) The two drift directions are independent: detuning s moves the constant along (9), sextic admixture r moves it along (7), and G^* is attained only at the corner $(0,0)$ of *these reduced* coordinates. In the bare parameters that corner requires $\mu/A^2 \to 0$, not merely $\mu \to 0$ and $A \to 0$: a path holding μ/A^2 fixed runs along a horizontal line of the plane and never reaches the corner. Contours are exact quadrature; the corner value agrees with $\sqrt{\pi}\,G^*$ to 10^{-9}.

The rate matters, and this is sharper than a double limit. Read back in the bare parameters, $\mu \to 0$ and $A \to 0$ *together do not suffice*: the reduced detuning s is a ratio, so the value attained depends on the path. Sending $\mu \to 0$ with μ/A^2 held at any nonzero constant leaves s fixed and returns $4\kappa K(\kappa)$ at that modulus, not $\sqrt{\pi}\,G^*$; taking $A \to 0$ first at fixed μ sends $s \to \infty$ and the constant to zero. Even the two iterated limits disagree. The condition that selects the lemniscatic value is $\mu/A^2 \to 0$ — the detuning must vanish *faster than the amplitude squared*. This is the price stated exactly: G^* is not merely a corner but a corner approached at a controlled rate. Figure 3 draws both statements at once: the one curve every detuning lies on, and the two-drift plane whose corner is the only point where the constant lives.

4.6 The native screen finds no quartic clock

Which cases does the model realize by itself? Its zero-tension equilibria are exactly the frameworks whose active bonds all sit at r_0 with opposite-polarity endpoints — unit-edge bipartite frameworks. A screen over every connected minimum-degree-two bipartite interaction graph up to $N = 7$, plus the cube graph, found **no self-stressed embedding**: seventeen graph classes, zero hits, with constraint-clean minima of $\lambda_{\min}(RR^{\mathsf{T}})$ bounded away from zero at 1.03×10^{-2}. Exactly, the cubic checkerboard blocks at $L = 2, 3, 4$ have self-stress dimension zero, so all their flexes extend. Larger stressed objects contract to networks of straight unit chains meeting at joints, and the two minimal joint graphs are squeezed by geometry: an interior four-joint configuration has a joint angle of at most 30° against a clearance requirement of 28.96°, and an exhaustive integer-distance search — 153,898 integral quadruples to diameter 60, 7,177 of them with an interior point — tops out at 26.14°.

One object survives the static gates: a nineteen-body hexagonal wheel, all twenty-four bonds exactly unit, with a one-dimensional equilibrium self-stress certificate (positive signs on the rim, negative signs on the spokes; not an applied force) and every individual chain-bowing flex blocked. It is, to our knowledge, the first zero-tension self-stressed equilibrium exhibited in this model. *And it fails as a clock.* Its quartic form is a squared difference, $E_4 \propto (\sum \kappa_{\text{rim}} - \sum \kappa_{\text{spoke}})^2$, so the combined flex that bows the rim and buckles the spokes equally costs nothing: the blocking form on the 28-dimensional nontrivial flex space is sign-indefinite, with eigenvalues from -1.467 to $+0.743$. Direct integration confirms it, the constrained static energy collapsing to $\sim 10^{-12}$. The verified per-chain quartic was a chord across a flat valley.

The failure sharpens the criterion, and the sharpened form is a genuine result of this paper:

Criterion 2 (Buckling)**.** *Chain stations are free hinges, so a* compressed *chain of length* ≥ 2 *buckles at zero energy; only tensioned chains are straight-stable. A native* $n = 4$ *clock therefore requires, beyond Criterion 1:* (1) *a self-stress;* (2) *a blocking form definite on the* whole *flex space, not merely mode by mode; and* (3) *every compressed member a* single bond*.*

Because an all-tension self-stress is impossible on a finite framework, condition (3) forces mixed structures — single-bond struts together with straight tensioned chains. Whether such an integer "unit-strut tensegrity" exists in this model is open; the first candidate family is killed exactly, its closure condition $s^2 - 4k^2 = -1$ being impossible modulo 4.

4.7 The minimum viable clock

Given the native screen's null, what is the cheapest purchased object whose leading soft mode realizes it? The minimal construction and its $A \to 0$ normal form are exact. Take four bodies $A^+B^-C^+D^-$ collinear at positions $0, 1, 2, 3$, with the three unit bonds at their minimum (curvature k_1) and *one additional interaction species* closing A to D at its own minimum at range $3r_0$ (curvature k_2). All four bonds are at zero tension. Then rank $R = 3$, the self-stress is one-dimensional with $\omega = (1, 1, 1, -1)$, and the blocking form for each transverse axis has profile spectrum $\{0, 0, 2, 10/3\}$, whose kernel is translation plus rigid rotation. After quotienting those trivial motions and including both transverse axes, the four-dimensional nontrivial flex space has spectrum $\{2, 2, 10/3, 10/3\}$ and zero kernel — positive definite on *every* nontrivial flex. The chain is second-order rigid at zero actual tension, certified by this positive self-stress form: $n = 4$ in every nontrivial infinitesimal-flex direction (longitudinal stretch directions remain quadratic). Crucially the certificate's one compressive member is a single bond, so Criterion 2 is satisfied where the wheel failed.

The mirror-even zero-momentum mode $q = (-1, +1, +1, -1)U$ is transverse, so it elongates the outer bonds only at *second* order and the chain relaxes longitudinally in response. Together with the two longitudinal stretches it drives, it spans a dynamically invariant *three*-degree-of-freedom sector; adiabatically eliminating the two stiff stretches closes it as a one-degree-of-freedom oscillator that is purely quartic in the $A \to 0$ limit — an instance of (5), with the blocked-flex coefficient evaluated on this stress — with

$$\lambda_{\text{eff}} = \frac{8k_1k_2}{k_1 + 3k_2}, \qquad m_{\text{eff}} = 4, \qquad \boxed{\lim_{A \to 0} T \cdot A = \sqrt{\pi}\, G^* \sqrt{\frac{m_{\text{eff}}}{2\lambda_{\text{eff}}}} = \sqrt{\pi}\, G^* \, \frac{\sqrt{k_1 + 3k_2}}{2\sqrt{k_1k_2}}} \qquad (10)$$

Direct integration of the ideal quartic normal form reproduces (10) to machine precision. Separately, an exploratory twelve-degree-of-freedom conservative *harmonic-spring surrogate* matched only at the stated curvatures holds $T \cdot A$ constant to 0.33% across a sixfold amplitude range, recovering G^* to 2×10^{-6} in the $A \to 0$ extrapolation *with no fitted scale* (Figure 4).

And the surrogate residual is computable, which is better than calling it small. Relaxing the harmonic-spring chain at fixed U gives a reduced system that is not the ideal normal form in two respects. Its potential carries a sixth order term,

$$E_{\text{eff}}(U) = \lambda_{\text{eff}} U^4 + \frac{16k_1k_2(k_1-k_2)}{(k_1+3k_2)^2} U^6 + \cdots,$$

which vanishes *only* at equal couplings — where the series runs $2U^4 - 2U^8 + 6U^{12}$, in powers of four alone — so the equal-coupling purity is a cancellation, not a law. And its reduced *mass* is amplitude dependent, $m_{\text{eff}}(U) = 4 + 4U^2 - \cdots$, a correction the sextic family of Section 4.5 does not contain at all. Quadrature on that surrogate reduction reproduces the simulated drift (+0.328% against +0.331% at $A = 0.12$) and decomposes it: +0.314% kinetic against +0.014% potential, a ratio of 22.7. The drift is therefore an *excess*, dominated by the mass term — which is why it is not the sextic admixture of Section 4.5, and why G^* is recovered only in the limit where both corrections vanish.

Minimality is exact by enumeration: at $N \leq 3$ the opposite-polarity graphs are trees and cannot carry stress; at $N = 4$ the single-scale embeddings have no stressed realization; and polarity alternation forces the closure span to be odd, hence $3r_0$ at minimum. **Four bodies and one additional interaction range is the floor.**

Two remarks about $\omega = (1, 1, 1, -1)$, since the sign pattern invites a reading it will not support. It is indexed by *bonds*, not directions, and its signs denote tension/compression capacity in an equilibrium self-stress certificate, not actual force at the potential minima: the lone minus sits on the closure bond, forced both by the impossibility of an all-tension self-stress on a finite framework and by a four-cycle having exactly one closure. The object that could carry a signature is the blocking form, whose coefficients these are — and it is positive *semi*-definite, spectrum $\{0, 0, 2, 10/3\}$, with no negative direction at all. Written in the consecutive stretches u_i it is $\propto 3\sum u_i^2 - (\sum u_i)^2$, the variance: the minus subtracts a mean and thereby removes the rigid-motion zero mode. It does not open a cone. An *indefinite* blocking form is precisely a buckling mode, which is how the wheel of Section 4.6 failed.

And the carrier cannot test what it was built beside. Recorded here, at the point of purchase, rather than where it is later needed: this clock is a mechanical framework, $m\ddot{q} = -\nabla V(|q_i - q_j|)$, and such an equation is *Galilean* invariant. Boosting it leaves the period exactly unchanged at any velocity, so it would report zero time dilation whatever the substrate did. The carrier is therefore disqualified as an instrument for the relativistic obligations of Section 9 — not by a defect in its construction, but by the kind of object it is.

The price. One additional interaction species, plus its scale parameters. What it buys exactly is a quartic leading normal form and the minimal four-body construction; the declared carrier is guaranteed only through its asymptotic normal form, and its compact-law finite-amplitude motion has not been simulated. Nor does the purchase by itself establish a spectral window. In the curvature-normalized harmonic surrogate used for the screen, $k_1 = k_2 = k$ and clearing the propagating band requires $kA_{\max}^2 > 1.0556$, or $k > 4.22$ at $A_{\max} = 0.5$. This number is not a compact-law depth bound: the registered unit bond has $k_1 = 96\varepsilon$ and a same-shape range-three closure has $k_2 = (32/3)\varepsilon_2$, so the conversion depends on the additional ratio ρ. A compact-law dynamical screen remains to be run. The carrier's true price therefore includes matter stiffness at field-stiffness parity. Its falsifier is stated in Appendix A.

4.8 What the lemniscatic price buys

The price is now fully stated: one interaction species, and a threshold that must be held. What has not been stated is what it returns. A ledger that records a cost and not its purchase is an unstable thing to leave standing, and the structure of the last three subsections settles it.

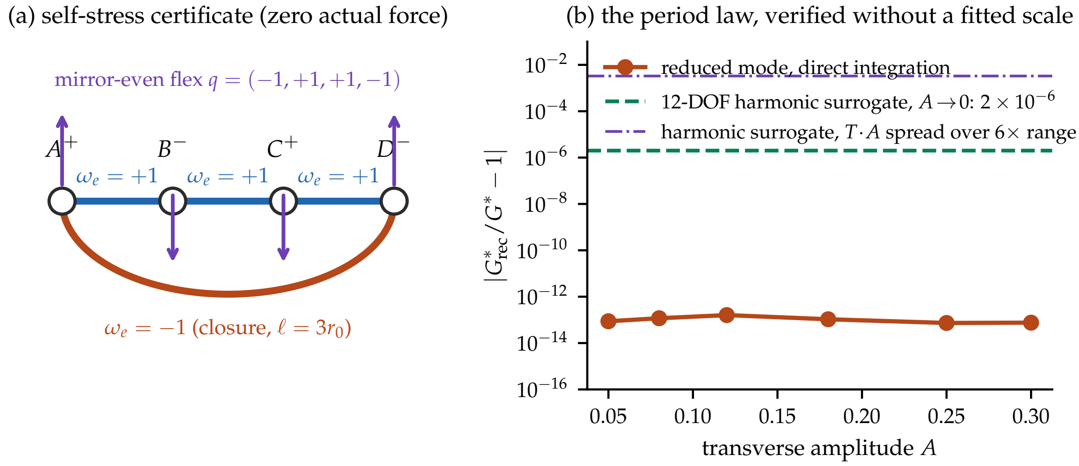

Figure 4: The minimum viable leading-order quartic clock carrier. (a) Four bodies, three unit bonds and one single-bond closure at range $3r_0$. The signs of $\omega = (1, 1, 1, -1)$ encode the possible equilibrium self-stress, not actual forces; the blocking form is positive definite on every nontrivial flex, so the unstressed framework is second-order rigid. (b) The $A \to 0$ period law (10) verified without a fitted scale: direct integration of the ideal quartic normal form reproduces it to machine precision, while the two horizontal lines mark the independent twelve-degree-of-freedom harmonic-spring surrogate and its finite-amplitude drift. They are not a simulation of the registered compact bond law.

Read the table of (4) again, this time as a list of degeneracies. Only $n = 2$ has nonzero second-order stiffness; every $n \geq 4$ is second-order *free* and still oscillates. Ordinarily those properties exclude one another — a stiff mode oscillates but resists being moved, a soft mode moves freely but does not oscillate — and the quartic is the *minimal* escape, needing exactly one vanishing Taylor coefficient where $n = 6$ needs two. G^* is the period constant of that minimal degeneracy.

The purchase is then exact. Fixing mass, displacement and period, the energy to hold the mode displaced is

$$\frac{E\,T^2}{mA^2} = 2\pi^2 \ \text{(harmonic)}, \qquad \frac{\pi}{2}G^{*2} \ \text{(quartic)}, \qquad \frac{E_{\text{quartic}}}{E_{\text{harmonic}}} = \frac{G^{*2}}{4\pi} = 0.6966, \tag{11}$$

the stiffness cancelling from both, so the ratio is pure G^*. And at displacement a the ratio is $(a/A)^2 G^{*2}/4\pi$, which vanishes: a degenerate direction is free to first order, so at a tenth of the reference displacement the quartic mode is 144 times cheaper to move. Figure 5 sets the two clocks beside each other — their constants as arc lengths, their orbits at equal energy, and the threshold surface on which the quartic one lives.

Proposition 1 located the genericity, and it is worth saying what that buys as well as what it costs. The quartic clock is the normal form of a perfect Euler column at critical load and of any mean-field symmetry-breaking order parameter; what is non-generic about it is not the shape of the potential but the *location*.

Why the maintenance is expensive, and not merely required. There is a sharper reason than codimension, and it is the same fact that made horology choose $n = 2$. A clock whose frequency tracks its own amplitude converts every amplitude perturbation into a frequency perturbation. Isochronism is not an aesthetic preference but *noise immunity*: the pendulum's $\partial T/\partial A \approx 0$ is the whole reason clocks work, and the anti-pendulum maximises exactly the coupling the pendulum was chosen to suppress. Near threshold the mode is soft, susceptibilities grow and relaxation slows — so the very sensitivity that makes critical points interesting makes them poor timekeepers.

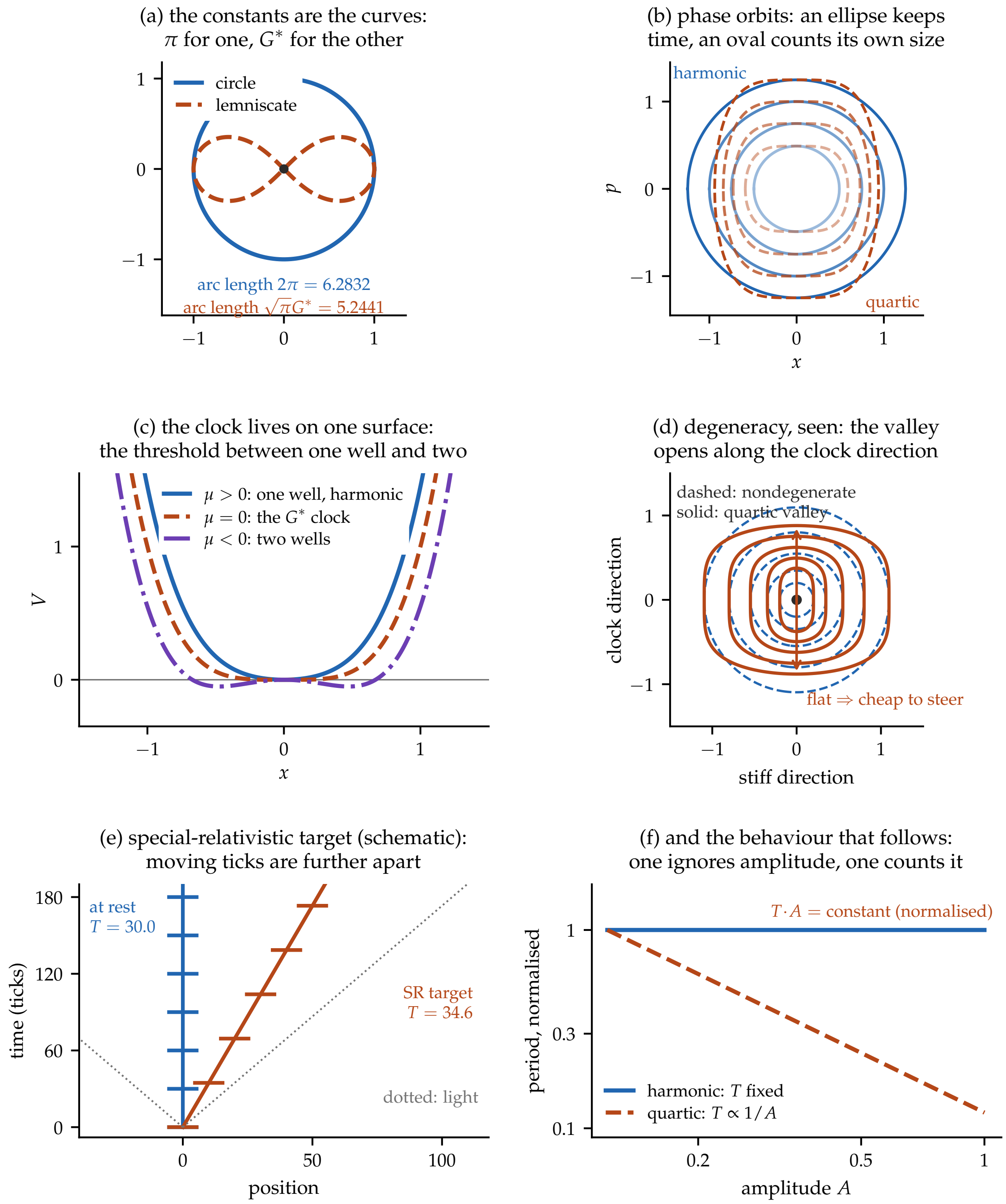

Figure 5: The two clocks, seen. (a) The constants are not normalisations but arc lengths: the unit circle's is $2\pi = 6.2832$, the unit lemniscate's is $\sqrt{\pi}G^* = 5.2441$ — the latter integrated numerically here and agreeing with $B(\frac{1}{4}, \frac{1}{2})$ to 10^{-7}. (b) Phase orbits at equal energy steps: harmonic level sets are ellipses traversed uniformly, so the period ignores the orbit's size; quartic ones are flat-flanked ovals and are traversed faster as they grow. (c) The degenerate clock does not live at a generic point but on a single surface — the threshold $\mu = 0$ between one well and two, where the cubic is forbidden by symmetry and the quartic is what remains. This is the picture of Proposition 1: the codimension-one surface is the hypothesis, and being off it is the generic case. (d) The degeneracy made visible: contours about a nondegenerate minimum are closed ellipses (dashed), while the quartic valley (solid) opens along the clock direction. That flatness is the cheapness of (11). (e) The special-relativistic dilation target as a schematic: tick events on two worldlines, one at rest and one at $u = 0.5\,C$, with the moving ticks separated by γ. This panel contains no soliton shape-mode measurement; that campaign is classified as unreproduced in Section 9.5.6. (f) The behaviour that follows.

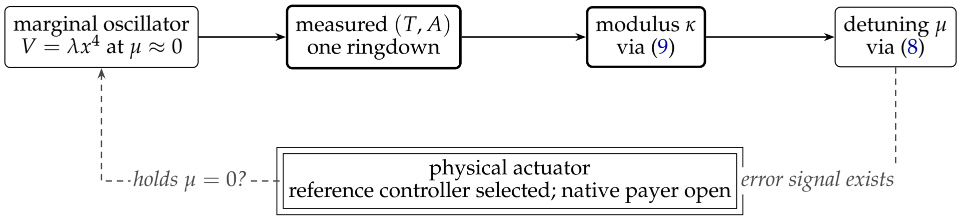

Figure 6: The sensor inversion and its status split. A marginal clock's timing fixes its own distance from the threshold — the solid path is exact — so the error signal needed to hold $\mu = 0$ demonstrably exists. The v2 reference controller consumes that signal with context-blind dual feedback and audits its work and dissipation. The double-ruled block is the still-missing native actuator: reference stabilization is not evidence that the substrate pays the maintenance cost.

That inverts usefully. A defect of a clock is a virtue of a sensor, and what this oscillator's timing is exquisitely sensitive to is *its own distance from the threshold*: by (9) the measured (T, A) pair fixes κ, and by (8) κ is the order parameter of the detuning. So a marginal clock reads out the quantity it must hold fixed. The v2 reference construction closes this as a control problem by selection: one context-blind feedback channel drives the dimensionless detuning $\delta = \mu/A^2$ toward zero, while a second stabilizes amplitude or action. Its audit records controller work, dissipation, detuning, and amplitude; maintenance is never treated as free. A gate opens only at a fixed phase crossing and under preregistered compliance tolerances. This constructs a mathematical controller, not a native substrate actuator. Figure 6 separates the implemented reference loop from that remaining physical debt.

So the price is a maintenance cost, continuously paid, not a construction cost; and (11) is what it returns. *π is native because sitting off the threshold is generic; G^* is priced because sitting on it must be actively held* — and what the holding buys is that timekeeping and steerability stop trading against each other. A carrier required only to be stable will be harmonic, since stability alone selects a point off the threshold; a carrier that must also be reconfigurable at low cost is pushed onto it.

This is also where the agency reading acquires a mechanism rather than an analogy. An agent, thermodynamically, is a system that spends free energy to hold a configuration against drift — and holding $\mu = 0$ is exactly that: an unmaintained system falls off the threshold and becomes a harmonic clock. The sharp form of the question is therefore not whether agents keep lemniscatic time, but that *a G^* clock is one which must be actively maintained, and what the maintenance buys is cheap actuation*. That is a claim about control cost. It neither requires nor supplies anything about consciousness, and marginal stability — a genuine hallmark of adaptive systems — is equally a property of sandpiles and buckling columns.

This interprets G^*; it does not recover it. The proposition forces the quartic *given* a threshold, a reflection symmetry and a nonvanishing quartic coefficient. Nothing here forces a substrate to sit at a threshold, and **no valid carrier has produced G^***: the native single-scale screen returned zero, the four-body carrier that does produce it needs a purchased interaction species and is Galilean and therefore disqualified as a relativistic instrument. The only positive dilation fit reported here is isochronous — a π-clock — and uses a fitted speed about six percent below the free-sector energy speed, so it does not establish a shared relativistic clock. G^* enters this framework by choice. Its selected role is purely temporal: it controls when a compliant event is eligible, while Σ controls which outcome occurs. Conflating those two roles would make the Born argument circular. Nor does the argument reach agency: it is a selection statement about efficient design, and marginal stability, while a hallmark of adaptive systems, is equally a property of sandpiles and buckling columns.

4.9 What a clock is owed by the rest of the paper

One forward debt is worth naming before leaving. The recursion this architecture runs on is indexed by a stage, and an index is not a duration: something physical has to carry it. That makes the clock not a subplot of the architecture but a precondition of its own bookkeeping. Section 10 pays this out, once there is a recursion on the page to be indexed.

A clock, though, is not yet a record. One that is reset by every fluctuation counts nothing, because counting requires that the last tick still be there when the next arrives. Persistence is a separate purchase and it is made in the same currency — an energy barrier — which is the subject of the next section.

5 Memory: the register

A clock alone carries no past. The minimal memory in the same model is a single body cradled by three anchors on an equilateral triangle of side $s < \sqrt{3}$: for such s there are two mirror equilibria at heights $\pm h$, $h = \sqrt{1 - s^2/3}$, with all three bonds at exactly r_0. The registered compact bond law is a polynomial in the squared normalized distance $q = (r/r_0)^2$:

$$V(q) = \begin{cases} -16\varepsilon(q - \frac{3}{2})^2(q - \frac{3}{4}), & q < \frac{3}{2}, \\ 0, & q \geq \frac{3}{2}. \end{cases} \tag{12}$$

It has $V(1) = -\varepsilon$ and radial curvature $\mathrm{d}^2V(r^2)/\mathrm{d}r^2|_{r=1} = 96\varepsilon$.

The barrier between them is not a fitted quantity, and it is not a measurement either. An explicit hinge path — break one bond, swing on the remaining two — costs exactly $1.000\,\varepsilon$, which bounds it above. The matching lower bound needs no grid over most of its domain.

Proposition 2 (Register barrier)**.** *For $s \in [0.9, 1.3]$ the register's barrier is exactly one bond's dissociation depth, independent of the geometry: $\Delta E = \varepsilon$, and the minimax level is attained by an in-plane two-bond configuration, one of the symmetry-related hinge crossings.*

Proof sketch. The two minima sit at $z = \pm h$, so any continuous path between them has a moment with $z = 0$: *every* path crosses the anchor plane. Hence $\Delta E \geq \min_{z=0} E - E_{\text{ground}}$, which replaces a min–max over paths by a minimisation over a plane. On that plane write N for the number of bonds whose squared distance lies in the attractive window $q \in (3/4, 3/2)$. The bond law satisfies $V \geq -\varepsilon$ everywhere, $V \geq 0$ for $q \leq 3/4$, and — decisively — $V \equiv 0$ for $q \geq 3/2$. So $E \geq -\varepsilon N$ pointwise, and wherever $N \leq 2$ we have $E \geq -2\varepsilon$. Outward-rounded interval arithmetic over boxes in (x, y, s) brackets every squared bond distance $q = d^2$ over the whole box. Over 11,250,000 closed boxes on an exact rational partition of $(x, y, s) \in [-3/2, 3/2]^2 \times [9/10, 13/10]$, every box has $N \leq 2$; there are zero three-attractive candidates and zero uncertified boxes. Every binary64 primitive is widened by one ULP with `nextafter`. Outside the square the origin bond has $q \geq 9/4 > 3/2$, so the exact $N \leq 2$ argument covers the rest of the infinite plane. No point samples or potential-energy fallback bounds enter the certificate. The value -2ε is attained: in-plane the body sits at distance exactly r_0 from two anchors, its distance to the third being $d_3(s) = s\sqrt{3}/2 + \sqrt{1 - s^2/4}$. Analytically, $d_3'(s) > 0$ whenever $s^2 < 3$, and $d_3(9/10) > 323/200 > 3/2$, hence $q_3 = d_3^2 > 3/2$; the third bond contributes exactly zero throughout the interval. With $E_{\text{ground}} = -3\varepsilon$ this gives $\Delta E \geq \varepsilon$, and the hinge path gives $\Delta E \leq \varepsilon$. □

Two holes in an earlier version of this certificate are worth recording, both found by review. It sampled five values of s rather than the interval and, more fundamentally, treated Euclidean distance d as the polynomial variable $q = d^2$. At $s = 1.3$ it compared the centroid distance $d = 0.75056$ with the squared-distance window edge $q = 3/4$, inventing a three-attractive pocket;

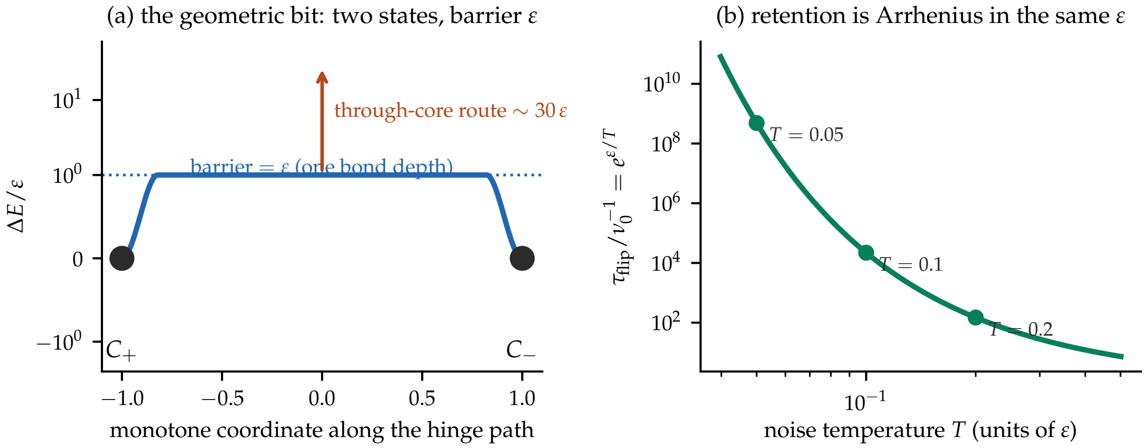

Figure 7: The geometric bit. (a) Two mirror states separated by a barrier of exactly one bond depth along the hinge path; the direct route through the repulsive core costs about thirty times as much. The matching lower bound is an exact plane-crossing argument plus an outward-rounded interval certificate with zero uncertified boxes. (b) The same ε enters the Kramers/Arrhenius retention estimate under its stated weak-noise and damping assumptions.

the correct centroid value is $q = s^2/3 = 0.56333\ldots < 3/4$, so no such pocket exists. Exact rational box coverage plus outward-rounded bounds on $q = d^2$ remove both errors. The certificate is machine-readable under schema `ftd.register_barrier.interval-certificate.v1`.

Compact support is load-bearing here, and worth naming as the modelling choice it is: under a Morse or Lennard-Jones tail the third bond would contribute a small negative amount at the crossing and the barrier would sit strictly *below* ε. The earlier watershed computation, which returned 1.016–1.021 ε, is now diagnosed rather than trusted — its 1.6–2.1% excess was entirely grid resolution, as the proof's exact 1.000 confirms. The direct route through the centre costs about 30ε.

Two structural observations follow, and the first is the reason this section sits beside the last. The register must be *rigid with a barrier*; the clock must be *flexible but blocked*. They occupy opposite corners of the very same rigidity criterion. And the same constant ε that sets the bond depth sets the retention time, through the Arrhenius form $\tau_{\mathrm{flip}} \sim \nu_0^{-1} e^{\varepsilon/T}$ when the imposed bath lies in the Kramers regime (Figure 7). Thus one constant prices the declared bond depth, register barrier, and retention exponent. It does *not* fix the clock rate by itself; Section 11 exposes the additional ratio $\rho = \varepsilon_2/\varepsilon$.

Both purchases so far were quoted in the same unit, ε, and neither section said what fixes it or what it costs to keep paying. That is a thermodynamic question — a barrier is only a barrier relative to a temperature — and it is the subject of the next section.

6 Thermodynamics: noninvertibility, retention, and timescale

Thermodynamics enters this architecture at three places. The distinctions below are deliberately narrower than the slogan that actualization "uses energy and cannot be undone." They separate a many-to-one record map, a metastable memory, and a complete thermodynamic entropy balance; only the first two are supplied here.

Singularity, readout loss, and full noninvertibility are different claims. Postulate 1 declares that one outcome enters history [IMPOSED]. Identifying the model's manifestation/genesis rule with that actualization block is a proposed physical identification [CONJECTURE]. For a fixed context M, write its discrete readout as

$$\mathcal{R}_M : X_M \longrightarrow O_M, \qquad x \longmapsto o. \tag{13}$$

The genesis readout is many-to-one: distinct continuous flux magnitudes can yield the same discrete record [MEASURED, engine-scoped]. The continuous flux itself remains stored, so this does *not* prove that genesis erases the complete microstate. Full-map noninjectivity is established separately by an evaporation witness: two distinct admissible engine states evolve to one identical persistent state [THEOREM, current-engine scope]. The honest statement is therefore that the candidate readout is lossy and the current complete update is noninjective, not that every manifestation event erases all information.

This separation also fixes the word *unchoose*. The discrete record does not reconstruct the continuous preimage or turn a counterfactual into the event that occurred. Reversing an enlarged microscopic state, if one exists, is a different operation from reading the inverse of (13).

What "energy-paid" means here. The accepted genesis map has a positive drain and is non-symplectic under the model's standard (J, W) canonical pairing. A branchwise symplectic, energy-conserving lift exists only after record-bearing bath variables and feedback are added [THEOREM, model-scoped]. At positive drain at least three canonical bath pairs are required for that selected pairing; after one event the bath is loaded and a fixed zero-bath section cannot be reused unchanged. Repeated operation therefore requires reset, replacement, export, or active feedback. One declared implementation test also removes about 37% of the field norm [MEASURED, implementation-scoped]. Neither number is a universal measurement-energy quantum or a complete heat/entropy balance.

The imposed bath is an operational thermostat. The stochastic element in the measurements of Section 8 is an Ornstein–Uhlenbeck thermostat on the field velocity, and it is not decorative: it equipartitions, delivering $\langle |v|^2 \rangle = 3T$ per site to within a few per cent, with an autocorrelation decaying at the declared rate. It is, however, *imposed*: the underlying dynamics is deterministic and supplies no ensemble of its own. Every probabilistic statement in this paper inherits that qualification.

Retention is Arrhenius; reset cost is a separate question. Proposition 2 put the barrier at one bond depth, and Section 5 gives $\tau_{\text{flip}} \sim \nu_0^{-1} e^{\varepsilon/T}$. A register is therefore not free: holding one bit for N clock cycles requires $\varepsilon/T \gtrsim \ln(N\,\nu_0 T_{\text{clock}})$, so the required barrier grows logarithmically with the desired retention depth — equivalently, achievable retention grows exponentially with ε/T. The model's statics supply the barrier value; the exponential form is imported Kramers rate theory under the imposed bath, and inherits its premises. This is a reliability bound, not an energy ledger for one selection. If a later reset logically erases one unbiased bit in an equilibrium bath, Landauer's bound supplies $W_{\text{reset}} \geq k_B T \ln 2$ [20]; no such reset cycle or heat flow is constructed here. Energy dissipated to an untracked environment cannot be recovered by simply reversing the retained record, but the amount dissipated per actualization remains **open**. Figure 8 states the resulting, narrower economy.

And a detector timescale. In the declared classical threshold instrument, the quantity controlling the interpolation from amplitude to action/occupation weighting is not an energy but a ratio of timescales: the mode frequency against the imposed noise correlation time. Calling that interpolation "quantum" would assume the Born pushforward that the paper leaves open.

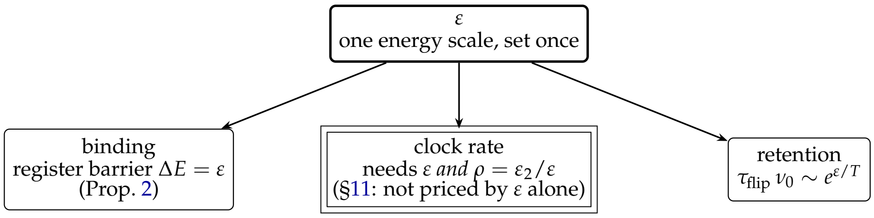

Figure 8: One constant, three bills — and the bill that did not clear. The same ε prices the matter's binding and the memory's lifetime, at exponents $+1$ and $+1$. It does *not* price the clock rate: the purchased closure species carries its own depth, so the clock needs $\rho = \varepsilon_2/\varepsilon$ as well, and ρ moves the clock while leaving the other two fixed. That is the architecture's own falsifier — a measurement moving one of the three without the others — and Section 11 fires it. The commitment survives for two of the three bills, and in the leading $A \to 0$ quartic normal form G^* itself is independent of ρ.

The debt this leaves. A clock prices duration; a barrier prices retention; the genesis analysis prices one model-scoped route to writing a record. None assigns the ensemble law of records, still less the joint law of responses under incompatible settings. A costly draw from a setting-independent marble bag is still a draw from the same bag. Thermodynamics answers how a fact can become stable. It does not answer why frequencies satisfy the trace rule or why incompatible contexts fail to glue into one Fine joint distribution. The next section imports the structure of quantum weights; the physical Born and Bell pushforwards will remain open after that import.

7 Weights, and where they come from

7.1 The adopted local net and the actual record algebra

For each finite region $\Lambda \subset \mathbb{Z}^3$, the actual beables are bounded functions of classical flux and ternary configurations,

$$\mathcal{A}_{\text{act}}(\Lambda) = \mathcal{B}_b\Big((\mathbb{R}^3 \times \{-1, 0, +1\})^\Lambda\Big). \tag{14}$$

This algebra is commutative. Its finite ternary record subalgebra is the diagonal algebra $\mathcal{D}_\Lambda \cong \mathbb{C}^{3^{|\Lambda|}}$; it is not the full matrix algebra $M_{3^{|\Lambda|}}(\mathbb{C})$. The potentiality layer is a separately adopted net $\Lambda \mapsto \mathcal{A}_{\text{pot}}(\Lambda)$ satisfying isotony, compatible restriction, and commutation for spacelike-disjoint regions. The first selected witness is

$$\mathcal{A}^{\text{ref}}_{\text{pot}}(\Lambda) = \bigotimes_{x \in \Lambda} M_3(\mathbb{C}). \tag{15}$$

Its finite-region algebras are Type I and its quasi-local inductive limit is UHF. No Type-III conclusion follows automatically: any factor classification in a GNS representation depends on the chosen state and representation, and finite random-matrix level statistics cannot decide it.

A preparation map Γ_Λ from substrate preparation classes to compatible local states ω_Λ is therefore an adoption with a named recovery debt. The compatibility condition $\omega_{\Lambda_2}|_{\mathcal{A}_{\text{pot}}(\Lambda_1)} = \omega_{\Lambda_1}$ for $\Lambda_1 \subseteq \Lambda_2$ is proved by the selected reference witness and must later be recovered physically. This is an abstract discrete local net first and a qutrit representation second; the classical ternary alphabet alone does not derive noncommutativity.

7.2 States, effects, and convexity

Two preparations are equivalent if every subsequent experiment gives the same statistics; the equivalence class is a *state* ω. Two detector outcomes are equivalent if they have the same probability for every preparation; the class is an *effect* e. The primitive pairing is

$$e(\omega) \in [0,1], \tag{16}$$

read as the potential weight of actualizing e given ω. There is as yet no Hilbert space, no wavefunction and no Born rule. If ω_1 is prepared with frequency p and ω_2 with frequency $1-p$, operational consistency forces $e(p\omega_1 + (1-p)\omega_2) = p\,e(\omega_1) + (1-p)\,e(\omega_2)$, so the state space is convex and effects are affine. This is the general-probabilistic level: it contains classical probability, real-Hilbert quantum theory, and much else.

7.3 The reconstruction, imported

What selects complex quantum theory from that class is a separate, external result. Chiribella, D'Ariano and Perinotti [1] derive finite-dimensional complex quantum theory from the operational-probabilistic framework together with six principles: causality, perfect distinguishability, ideal compression, local distinguishability, pure conditioning, and purification — the last being what separates the quantum member from the classical one. Under those premises,

$$\mathcal{H}_A \simeq \mathbb{C}^{d_A}. \tag{17}$$

Status of (17). A theorem *external* to this paper, conditional on its six imported premises. Nothing here derives those premises from anything more primitive; Appendix B itemizes them as an import invoice, which is the honest replacement for treating "the Hilbert space" as a single unpriced assumption.

7.4 The trace rule from effect additivity

Once the quantum effect space is in hand, the shape of the weight functional is fixed by a second declared postulate.

Postulate 2 (Effect-noncontextual additivity)**.** A weight functional $w : \mathcal{E}(\mathcal{H}) \to [0,1]$ with $w(I) = 1$ satisfies $w(E+F) = w(E) + w(F)$ whenever $E + F \leq I$, for *all* effect decompositions, including those whose members do not commute.

Postulate 2 is substantive and load-bearing: it is a noncontextuality of valuation over generalized measurements, and a contextual valuation escapes what follows without contradiction. From additivity, $w(qE) = q\,w(E)$ for rational q. Monotonicity is derived rather than assumed — for effects $E \leq F$ the difference $F - E$ is an effect, so $w(F) = w(E) + w(F-E) \geq w(E)$ — and squeezing with rationals $q_n \uparrow \lambda \leq r_n \downarrow \lambda$ gives $w(\lambda E) = \lambda w(E)$ for real λ. Extending by differences yields a positive linear functional, and in finite dimension every such functional is a trace:

$$w(E) = \mathrm{Tr}(\rho_w E). \tag{18}$$

This is Busch's generalized Gleason theorem [2], valid already in dimension two, where the original Gleason theorem — which assumes additivity only over *projective* decompositions — requires $\dim \geq 3$ and fails. The strengthening from projectors to effects is exactly the content of Postulate 2. Throughout this paper, (18) and everything downstream of it is described as *derived from Postulate 2*, never as derived *simpliciter*.

7.5 Probability is state plus question

With (18) the logical order of Section 2 becomes formal. For a measurement $M = \{E_i\}$ with $\sum_i E_i = I$,

$$p(i \mid \rho, M) = \mathrm{Tr}(\rho E_i), \qquad \sum_i p_i = \mathrm{Tr}\,\rho = 1, \tag{19}$$

so a probability exists only once a question is supplied: the state is not itself a probability distribution. For a pure state and a sharp outcome this reduces to the familiar square,

$$p_i = \mathrm{Tr}\left(|\psi\rangle\langle\psi|\,|i\rangle\langle i|\right) = |\langle i|\psi\rangle|^2, \tag{20}$$

which in this architecture is a *theorem about weights* standing at the end of the chain $|\psi\rangle \to \rho \to (\rho, M) \to p_i$, rather than an axiom about probability standing at its beginning.

Two structural facts follow immediately and matter later. Global phase is unphysical, so pure potentiality is a ray and ψ decomposes into a weight structure and a phase structure, of which squaring retains only the first. And coherent potentiality is not ignorance over the same alternatives: for $|\psi\rangle = \alpha|A\rangle + \beta|B\rangle$ the cross term $2\,\mathrm{Re}\,\alpha^*\beta\langle A|E|B\rangle$ is absent from the corresponding mixture. That argument excludes one specific mixture reading; the stronger exclusion of ψ-epistemic models is the Pusey–Barrett–Rudolph theorem [5], at the price of its preparation-independence assumption.

7.6 Dynamics, instruments, and the selector

Requiring affineness on mixtures, positivity, and positivity under extension by an untouched ancilla gives complete positivity, so physical channels are CPTP maps in the instrument tradition of Davies and Lewis [4]. A reversible transformation preserving pure-state transition probabilities acts on rays and is unitary or antiunitary (Wigner [6]); for a one-parameter group the antiunitary branch is excluded because $U(t) = U(t/2)U(t/2)$ and the square of either is unitary, and the projective obstruction vanishes for $\mathbb{R}$ [7], so with strong continuity Stone's theorem [8] gives $U(t) = e^{-itG}$ and, on defining $H := \hbar G$,

$$i\hbar\,\partial_t|\psi\rangle = H|\psi\rangle, \qquad i\hbar\,\dot{\rho} = [H, \rho]. \tag{21}$$

The existence of a phase generator is structural; the numerical value of $\hbar$ is empirical and is not derived here. A measurement context is represented not by an observable but by an *instrument* $\{\mathcal{I}_o^M\}$ of CP trace-nonincreasing maps summing to a trace-preserving map, with $E_o^M = (\mathcal{I}_o^M)^*(I)$; an instrument carries two things at once, the weight of a possible actualization and the successor potentiality if it occurs.

What the state/effect and instrument architecture does not supply intrinsically is the selection of one outcome. Writing Σ_O for that block, there are two mathematically distinct implementations,

$$\underbrace{K_O(o \mid \rho, M, H)}_{\text{primitive stochastic}} \qquad \text{or} \qquad \underbrace{\lambda \sim \mu(\mathrm{d}\lambda \mid \rho, M, H), \;\; o = F_O(\rho, M, H, \lambda)}_{\text{context-complete deterministic}},$$

and in either case the physical content is the requirement that the resulting frequencies reproduce the weights,

$$(F_O)_*\mu(\{o\}) = \mathrm{Tr}(\rho E_o^M), \tag{22}$$

called the *Born pushforward* below. V2 supplies the following **[IMPOSED reference realization]**. For the complete joint context C, order its outcomes and write $w_o^C = \omega(E_o^C)$. With $u \in [0,1)$ distributed by Lebesgue measure, set

$$\Sigma_C(u) = \min\left\{o : u < \sum_{j \le o} w_j^C\right\}, \qquad u_{n+1} = 2u_n \bmod 1. \tag{23}$$

Lebesgue measure is invariant under doubling. The quantile intervals have exactly the required lengths:

$$(\Sigma_C)_*\mu_{\mathrm{eq}}(o) = \omega(E_o^C). \tag{24}$$

The hidden measure is context independent, $\mu_{\text{eq}}(\lambda \mid C) = \mu_{\text{eq}}(\lambda)$, while the response map depends on the complete context and is intentionally nonfactorizable across Bell wings. Thus the construction proves deterministic contextuality, measurement independence, and compatibility with the target weights. It is *not* a Born derivation, because those weights are read to place the quantile boundaries. The physical programme must recover both the equilibrium measure and the pushforward from substrate dynamics without target-coded weights; its first preregistered candidate is the free-flux positive-frequency density $n = |\phi_+|^2$.

Note that Σ_O means the whole implementation — the kernel, or the pair (F_O, μ) — so that the measure cannot be hidden inside a deterministic map. Given an outcome, the successor state is ordinary conditionalization, $\rho' = \mathcal{I}_k^M(\rho)/p_k$; histories compose by $P(h) = \mathrm{Tr}[\mathcal{I}_{o_N}^{M_N} \circ \cdots \circ \mathcal{I}_{o_1}^{M_1}(\rho_0)]$, so the potential histories are plural while the actual history is singular. Collecting the pieces gives the recursion of Figure 1,

$$(H_n, \mathcal{C}_n, \mathcal{D}) \xrightarrow{\Gamma} \rho_n \longrightarrow \{(e_i, w_i)\} \xrightarrow{K \text{ or } (F,\mu)} e_{n+1} \longrightarrow H_{n+1}, \tag{25}$$

in which $\mathcal{C}_n$ is the preparation and control record and $\mathcal{D}$ the stipulated dynamics. The closure map Γ is part of the scaffold, not a theorem: history alone does not determine a state.

The weight layer is now assembled, at a cost stated line by line. The abstract local-net axioms and the finite qutrit witness are separate adoptions; the latter is Type I, its quasi-local limit is UHF, and any non-Type-I GNS factor would require an additional state-and-representation argument. But even granting the adopted net, nothing in this layer says physically *when* a weight becomes a frequency. That is not a question about the algebra of weights; it is a question about the mechanism that reads them out, and about how fast it reads compared with how fast the thing being read is moving. The next section measures exactly that ratio.

8 Action/occupation weighting as a detector regime

8.1 What a threshold counts

Consider the simplest possible actualization mechanism: a coherent field riding on noise, and an event recorded whenever the total crosses a threshold. Which feature of a mode does such a counter weight?

There are three candidates with different physical meanings. It might weight by *amplitude squared*, which is what a naive excess-rate calculation gives at leading order. It might weight by *energy*, which carries an extra factor of Ω^2. Or it might weight by *action/occupation* $n = E/\Omega$ — amplitude squared times frequency, the classical action variable and the adiabatic invariant. This endpoint has the same frequency dependence as mode occupation after the usual relativistic normalization, but that algebraic resemblance does not make it a quantum probability. The mechanism-level intuition is narrower: a threshold rate can factor into presentations per unit time and success per presentation. The deterministic traces below establish the frequency count and a binary amplitude gate, but do *not* derive a success law proportional to A^2 or their product as occupation. The noisy interpolation is measured separately. Whether any substrate turns that classical count into $\mathrm{Tr}(\rho E)$ is the separate Born-pushforward problem.

Figure 9 draws the counting itself, deliberately without noise: two modes a threshold cannot tell apart by occupation — one tall and slow, one half the height knocking four times as often — and two limiting faces of thresholding: presentations counted at a low level and amplitude gating at a high one.

The three classical weighting hypotheses are distinguishable. Preparing two standing modes at *equal occupation* and regressing the excess crossing rate on their spatial signatures

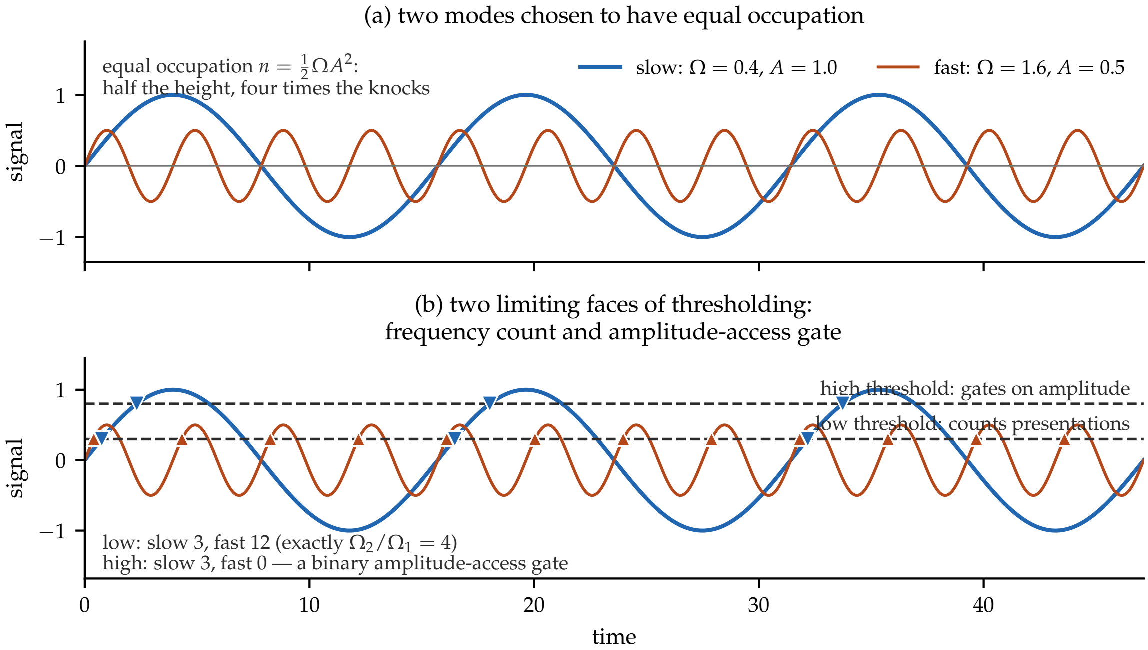

Figure 9: The mechanism, before any measurement. (a) Two modes at equal occupation $n = \frac{1}{2}\Omega A^2$: the fast one is half the height and presents four times as often. (b) The same traces against two thresholds, every crossing computed and marked. At the low threshold both modes clear, and the counts are exactly in the frequency ratio (12 against 3): crossings count *presentations*. At the high threshold only the tall mode reaches, and the fast mode logs zero: *success* gates on amplitude. A noisy threshold samples between these faces, and the interpolation between them is measured separately. The hard-threshold trace does not establish $p_{\text{success}} \propto A^2$ and therefore does not by itself derive occupation weighting. This figure is an illustration on one declared trace, not a measurement; the measured interpolation between the two weightings is Figure 10.

yields a ratio R whose value is Ω_1/Ω_2 under amplitude weighting, 1 under occupation weighting, and Ω_2/Ω_1 under energy weighting. We report the normalized interpolation coordinate

$$\text{OF} = \frac{R - R_{\text{amp}}}{R_{\text{occ}} - R_{\text{amp}}}, \qquad R_{\text{occ}} = 1, \tag{26}$$

the *occupation fraction*. It equals 0 at the amplitude endpoint and 1 at the occupation endpoint. It is a regression coordinate, not an event probability.

8.2 The measured interpolation

Under the author-attested locked protocol — outcomes, gates, kill conditions and runner hashes recorded as described in Appendix A — the answer is neither of the three fixed alternatives. The weighting *interpolates*, and the control parameter is the product of mode frequency and noise correlation time. Across five arms spanning $\bar{\Omega}\tau$ from 0.64 to 78.36 the occupation fraction rises strictly monotonically from 0.049 to 0.836 (Table 1, Figure 10), with a descriptive crossover near $\bar{\Omega}\tau \approx 17$.
The reading, at exactly its scope: *within the declared mechanism class, action/occupation weighting is approached in the fast-mode/slow-noise regime of a classical threshold statistic*. This is neither an exact asymptotic law nor a theorem about any substrate; whether the occupation fraction tends to exactly 1 or near the descriptive 0.86 is open. Identifying the endpoint with quantum outcome probabilities would require the still-open pushforward (22).

Instrument	$\bar{\Omega}\tau$	occupation fraction	99% CI
saturation scan, arm 1	0.64	0.0494	[0.0359, 0.0623]
saturation scan, arm 2	2.54	0.1020	[0.0898, 0.1144]
saturation scan, arm 3	10.15	0.2735	[0.2549, 0.2903]
saturation scan, arm 4	19.59	0.4694	[0.4583, 0.4797]
saturation scan, arm 5	78.36	0.8362	[0.8130, 0.8612]
earlier discrimination, slow	≈ 0.7	0.024	[0.014, 0.036]
earlier discrimination, fast	≈ 2.7	0.099	[0.090, 0.109]
slow-channel test, $f = 1$	141.7	0.9361	[0.8953, 0.9700]

Table 1: Occupation fraction (26) against the frequency–correlation-time product. The two instruments are reported separately and are *not* averaged: the slow points differ by about 0.02 beyond seed noise, a declared phase-draw systematic. The last row is a pure slow channel rather than a sixth arm of the same scan.

8.3 The negative, reported at equal prominence

The same law was then tested where it would matter more: on the model's own thermal field, with its own propagating noise, under a locked ten-cell design. **It did not transfer.** No declared axis reached the registered structure threshold — the Spearman correlation of occupation fraction against the primary axis was $+0.01$, against the secondary $+0.37$, and against threshold height -0.20, all far below the required 0.7. The registered outcome is negative.

The post-hoc diagnosis — and it is flagged as post-hoc, not registered — is visible in Figure 11: the mean-subtracted noise correlation time sat pinned between 0.6 and 1.9 ticks in every cell regardless of the thermostat's damping, so the accessible range of $\bar{\Omega}\tau$ was only 0.19–0.86, entirely below the crossover. Within that range the measured values, 0.01–0.05, are quantitatively consistent with the interpolation law. The structural suggestion is that thermal noise carried by the field itself can never be slow relative to that field's own band, because signal and noise are the same medium. We state that as a suggestion: the record disfavours the route in the sampled domain and does not prove that it can never succeed.

8.4 A slow channel, and the purity requirement

That diagnosis has a testable consequence: if the gate must be slow and the field cannot be, the model must be searched for a slow channel of different origin. It has one — a slowly accumulating potential sector sourced by the matter configuration. Measured under its own locked pre-registration on a self-gravitating configuration with no thermostat and no manifestation events, its fluctuations have correlation time $\tau = 317.9$ ticks against the field's 0.81: a ratio of 391, and $\bar{\Omega}\tau = 141.7$, deep in the regime the interpolation law identifies.

Feeding a channel at that timescale into the mechanism, with the fast field noise still present at full strength, gives the result in Figure 12: the occupation fraction stays pinned between 0.04 and 0.09 up to a slow-channel share of 75%, and reaches 0.936 only when the slow channel is essentially the *sole* stochastic element.

> **Measured purity diagnostic [MEASURED — instrument-scoped].** *In the additive two-channel model class studied here*, crossing statistics are dominated by the fastest stochastic component present, so occupation weighting requires not merely the presence of a slow channel but its near-exclusivity at the threshold. The evidential base is five slow-channel shares at 32 seeds on one instrument. This is an empirical diagnostic for that class, not a proposition about crossing statistics in general.

This is a demanding condition within the declared additive two-channel mechanism class: a

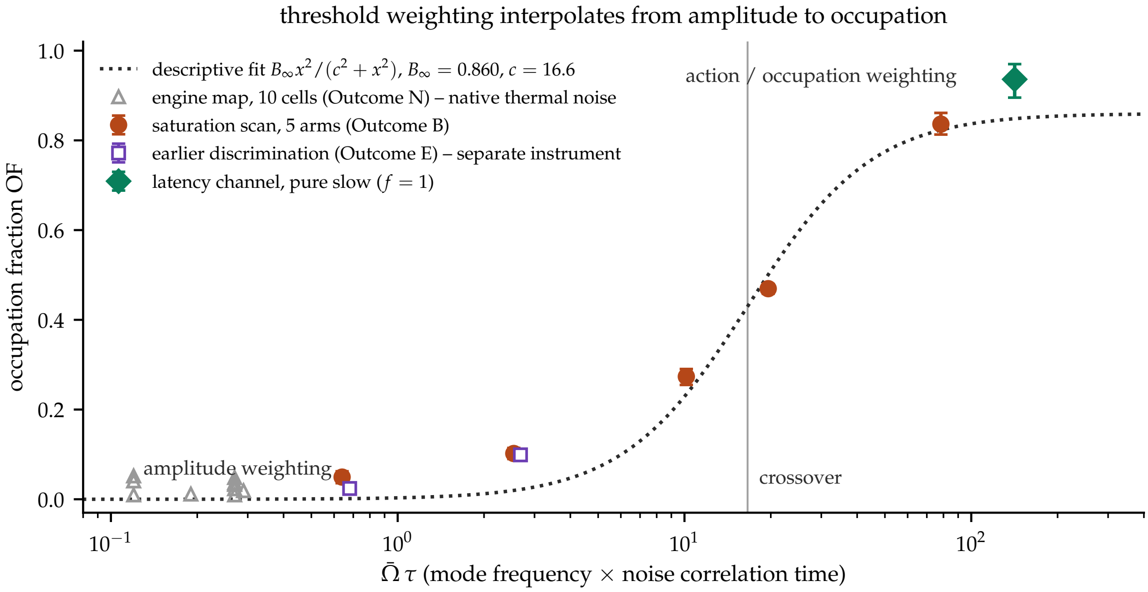

Figure 10: The classical detector measurement. Threshold-crossing statistics interpolate from amplitude weighting toward action/occupation weighting as mode frequency exceeds the noise bandwidth. Markers distinguish instruments deliberately: the two slow-point estimates are never averaged, the engine cells are a different noise structure spanning only $\bar{\Omega}\tau = 0.19$–0.86, and the highest point is a pure slow channel. The dotted curve is descriptive and carries no decision weight.

candidate realization in that class would have to *shield* the actualization comparison from the fast field, not merely add a slow term to it. No such necessity has been established for other detector dynamics.

8.5 What an irreversible choice does — and does not — buy

The marble example of Section 2 captures two real features of this ontology. Only the marble actually drawn enters the record, and a physical draw plus reset can be hysteretic and dissipative. The unrealized draws remain counterfactual. Neither fact, however, removes Bell's constraint. Bell does not require all four settings to be performed in one trial, nor does it require the record-forming dynamics to be reversible. It asks whether the four possible responses can be represented together on one setting-independent probability space.

Proposition 3 (Irreversible choice is Bell-neutral)**.** *Let $x, y \in \{0,1\}$ be the settings, let $\lambda \sim \mu$ be independent of their choice, and suppose the possible local responses are four functions $A_x(\lambda), B_y(\lambda) \in \{-1,+1\}$ on that one space. After those signs are selected, allow an arbitrary* outcome-preserving *physical write D_{xy} — including a many-to-one, hysteretic, or energy-dissipating update of the memory's microstate. Then correlators of the recorded signs obey $|S| \le 2$. Noninvertibility of that write does not weaken the bound.*

Proof. For every λ before the record update,

$$C(\lambda) = A_0(B_0 + B_1) + A_1(B_0 - B_1) = \pm 2,$$

because one of $B_0 + B_1$ and $B_0 - B_1$ vanishes and the other is ± 2. Averaging over the single setting-independent μ gives $|S| = |\int C \, d\mu| \le 2$. Any local stochasticity can be included in λ. The later map D_{xy} never enters the pointwise identity, so its invertibility and energetic cost are irrelevant to the inequality. If a write locally recodes a sign, that recoding can be

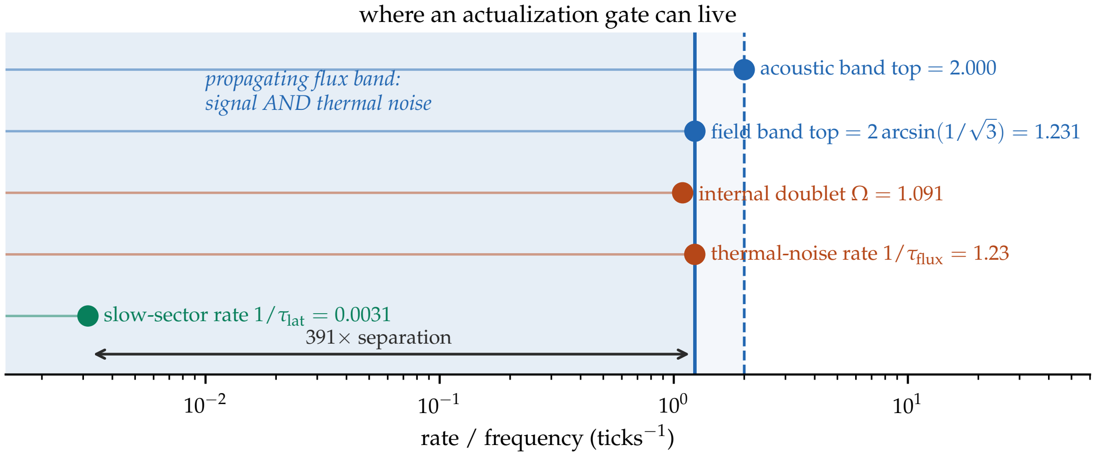

Figure 11: Why the negative may reflect a shared-band limitation in the sampled design. The propagating band carries both the signal modes and the thermal noise; in these ten cells the noise rate was not pushed below the band. A slower channel of different origin — here the model's slowly accumulating gravitational-potential sector, three orders of magnitude slower — is therefore a candidate location for an actualization gate, not a demonstrated necessity.

absorbed into the corresponding response function. A joint setting-dependent overwrite is not a counterexample; it abandons the local-response premise. □

The proposition marks the exact boundary of the user's intuition. An actual choice can be one-way without the unchosen alternatives being co-actual quantum worlds; that is a coherent ontology of singular records. But if the bag's marble positions and drawing dynamics define all four responses on one common ensemble, it remains a classical Bell model. Fine's theorem makes the joint-distribution boundary exact [3]. To reach $2\sqrt{2}$, at least one named premise must fail. V2 keeps measurement independence and ordinary equilibrium probability, but rejects the existence of four context-free local response variables: Σ_C depends on the complete joint context and does not factorize across the wings. This is an explicit ontic cost, not an evasion produced by discreteness. The imposed reference selector reproduces the selected singlet law while keeping each local marginal independent of the remote setting. Energy-paid record formation still explains only why an outcome persists; it does not derive the physical quantum correlation law from substrate dynamics.

8.6 Three ceilings, and the gap nobody has closed

The architecture's central distinction has so far been qualitative: potentiality is *structured*, and actuality is singular. Bell correlations make the first half quantitative, and doing so exposes an open problem that is the paper's thesis in numerical form.

Three ceilings bound the CHSH quantity, and each corresponds to a different admissible explanatory structure:

ceiling	value	structure that could produce it
local	2	an ensemble of pre-assigned values
quantum (Tsirelson [37])	$2\sqrt{2} \approx 2.8284$	a noncommutative weight structure
no-signalling (PR [38])	4	a nonsignalling conditional probability assignment

Three precisions keep the reading honest.

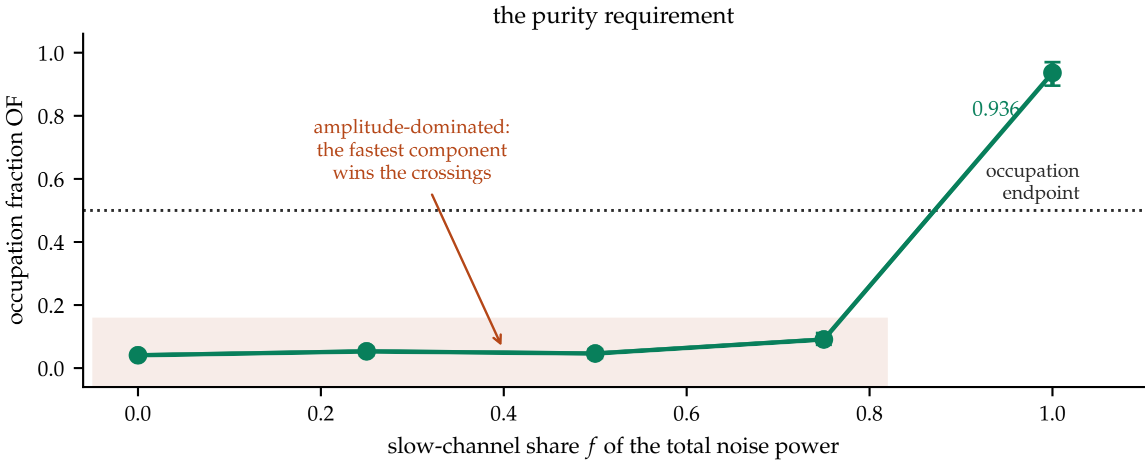

Figure 12: The purity requirement within the declared additive instrument. Mixing a slow channel into fast noise does not gradually produce action/occupation weighting; the classical threshold statistic remains amplitude-dominated until the slow channel is nearly exclusive.

The first ceiling is not a bound on actuality. Every recorded outcome is actual whatever S turns out to be; actuality is singular per trial and is not bounded by anything. What 2 bounds is what *pre-assigned* values could produce — the sixteen vertices of the local polytope are exactly the deterministic assignments in which each system already carries an answer to both questions it might be asked. This is why singular actuality and $S = 2\sqrt{2}$ coexist without strain: the tension is with pre-assignment, not with actualization.

The second is a slice, not a shape. $2\sqrt{2}$ is the maximal value in the CHSH direction of the quantum set. That set is nonpolytopal; general membership admits systematic outer bounds through the NPA semidefinite hierarchy [39].

The third is not merely a formal maximum. Popescu–Rohrlich boxes attain 4 while remaining operationally nonsignalling [38]. This does not establish compatibility with the causal structure of local QFT or AQFT. The outer ceiling is what nonsignalling probabilistic consistency alone permits.

That last point is the substance. **No-signalling and probabilistic consistency alone do not explain why nature stops at $2\sqrt{2}$.** The no-signalling set extends to 4; the quantum ceiling uses 70.7% of that range and leaves the rest excluded. Several principles have been proposed that recover Tsirelson's bound or approach it — information causality [40], macroscopic locality [41], local orthogonality [42], and limits on communication complexity [43] — and none is established as *the* derivation.

Figure 13 computes all three ceilings, and adds what the number buys. Device-independent certification turns S into a resource: at $S = 2$ the outcomes could have been written down in advance and the certified min-entropy is exactly zero, while at $2\sqrt{2}$ it is exactly one bit. Whatever fixes the ceiling therefore fixes how much unpredictability the world makes available — a physical quantity, not a bookkeeping one.

Within the adopted commuting-wing operator model, the usual CHSH square identity gives $\|\mathcal{B}\| \leq 2\sqrt{2}$. Consequently a PR-box value of 4 is rejected inside this reference architecture, even though it satisfies no-signalling. The selected singlet state and observables saturating the quantum ceiling remain imported reference data, not substrate outputs.

The architecture therefore acquires a quantitative form and an open problem in the same stroke. The potential is not "whatever is not forbidden": it is a specific, tightly constrained subset of the consistent, and the constraint has no accepted explanation. Whatever fixes the weight structure of Section 7 is the same thing that fixes this ceiling, and neither the

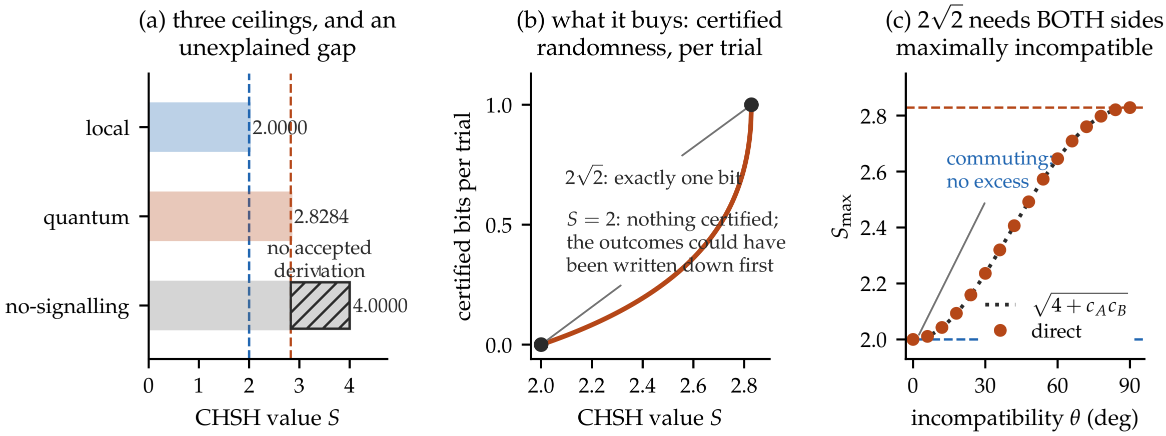

Figure 13: The three ceilings, computed. (a) Local correlations stop at 2, quantum at $2\sqrt{2}$, and no-signalling at 4; the hatched band is the 29.3% of the no-signalling range above the quantum ceiling. (b) What the number buys: the device-independent min-entropy runs from exactly zero certified bits at $S = 2$ — where the outcomes could have been written down in advance — to exactly one bit at $2\sqrt{2}$. (c) The operator-norm ceiling for dichotomic observables obeys $S^2_{\max} = 4 + \|[A_0, A_1]\|\,\|[B_0, B_1]\|$: each commutator norm is capped at 2, so a product of 4 requires both at maximum. There is no partial-credit route to $2\sqrt{2}$. The dotted curve is the identity and the markers are the directly computed operator norm; they agree to 2×10^{-15}.

reconstruction programme nor anything in this paper supplies it. The gap between $2\sqrt{2}$ and 4 is the width of that ignorance, measured.

8.7 What the angles do, and which part needs a record

One further separation is worth making explicit, because two claims are commonly run together and only one of them is true.

The CHSH number is a construction. Its four correlators are measured on four disjoint sub-ensembles; no trial contributes to more than one; and no individual context is anomalous — each sits at $1/\sqrt{2}$, comfortably inside the unit bound available to any correlator whatever. S is assembled, and it does not exist without something that assembles it. This is Fine's gluing statement in operational dress [3], and it is correct.

Order dependence is not CHSH compilation. Two crossed polarisers pass nothing; insert a third at 45° and $I_0/8$ passes. That sequential projection is reproduced by classical Jones/Maxwell optics as well as by quantum optics, so it is not by itself evidence for uniquely quantum observable noncommutativity. It does show that an ordered physical process need not wait for four-context statistical compilation. Quantum noncommutativity has independent consequences. For the canonical harmonic oscillator, $[x, p] = i\hbar$ together with $H = p^2/(2m) + m\omega^2 x^2/2$ yields $E_0 = \hbar\omega/2$; noncommuting dichotomic observables enter the CHSH operator-norm bound below.

The distinction matters for the architecture. The commutator is the *input* to $S^2_{\max} = 4 + \|[A_0, A_1]\|\|[B_0, B_1]\|$ — an identity for the operator norm, and a *bound* on any state's observed $\langle S \rangle$ — and that observed value is the *readout* (Figure 14). Records are required to estimate the correlators and compile the observed S, while the abstract operator algebra fixes the quantum ceiling before that compilation. This separates the algebraic premise from its statistical readout; the three-polariser example illustrates order dependence but does not decide the ontology of the quantum commutator.

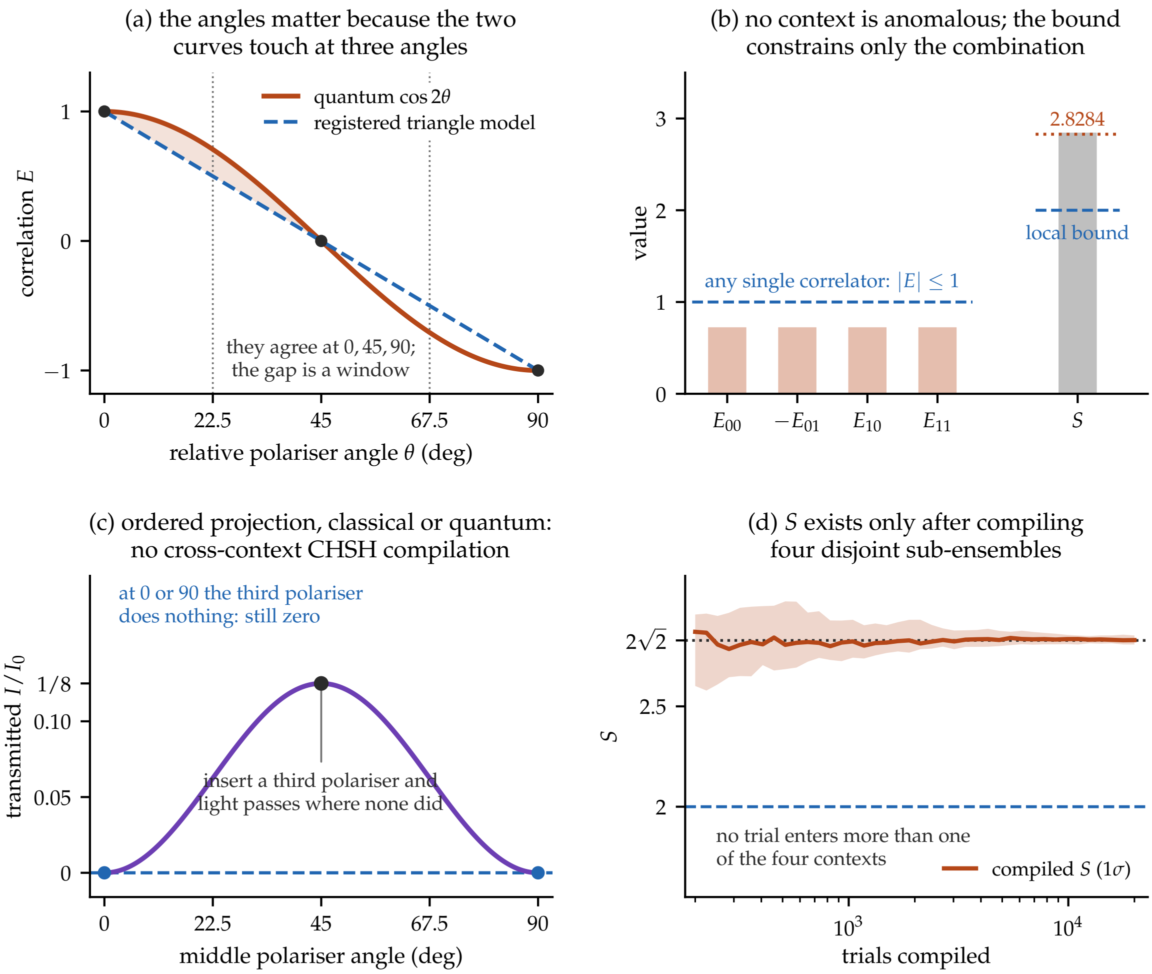

Figure 14: The angles, and the boundary. (a) For polarisation-entangled light the quantum correlation $\cos 2\theta$ and the registered deterministic sign-threshold triangle model *touch* at 0°, 45° and 90°; analytically the largest positive separation from that model is 0.210514 at 19.7701°. This line is not a universal pointwise locality bound; locality constrains the cross-context CHSH combination. (b) At the CHSH settings each of the four contexts contributes exactly $1/\sqrt{2}$, well inside the $|E| \leq 1$ available to any correlator: no context is individually anomalous, and the local bound constrains only the total. (c) Two crossed polarisers transmit nothing; inserting a third at 45° transmits $I_0/8$. Classical Jones/Maxwell optics and quantum single-photon optics both reproduce this order-dependent projection. It needs an output readout, but no cross-context CHSH compilation. (d) S by contrast is a compiled statistic: the four sub-ensembles are verified disjoint and exhaustive, no trial enters more than one, and the number exists only after all four are combined.

Scope of Section 8. All ensembles are imposed; the mechanism-level measurements run on a standalone instrument rather than the full model; the coupling in the last experiment is a declared additive model class. The slow-channel measurement's profile contains no random-number consumer, so its run is fully deterministic and its seed-independence is vacuous — configuration-varied robustness is a registered refinement, not a completed check. None of this establishes a Born law for any substrate. The tier structure of Section 8.6 is a restatement of established bounds, not a new result; its contribution is to locate the paper's open question inside them.

Everything established to this point is a *one-frame* statement: a regime condition read on one instrument, a ceiling fixed by one algebra, a clock ticking where it was built. Relativity is the demand that two frames agree, and it is the first obligation in this paper that cannot be discharged by any single apparatus, however good.

9 Relativity

The architecture divides cleanly under relativity, and the division is the paper's organizing claim about what merges and what does not.

9.1 The weight layer relativizes; this is standard physics

Covariant potentiality is the assignment of local algebras to spacetime regions with Einstein causality, $[\mathcal{A}(O_A), \mathcal{A}(O_B)] = 0$ for spacelike-separated regions [14, 15], with probabilities given by the state–effect pairing $p_\omega(E) = \omega(E)$ — the representation-independent replacement for the trace. Continuum AQFT local algebras may be non-Type-I and admit no intrinsic density matrix; this is an external comparator, not a classification of the finite qutrit net in (15). Covariant evolution of the potential layer is the Tomonaga–Schwinger equation [17], conditional on integrability. And for causally disjoint local instruments the joint statistics are independent of composition order,

$$p_\omega^{A\prec B}(a, b|x, y) = p_\omega^{B\prec A}(a, b|x, y), \tag{27}$$

provided the instruments satisfy a causal local measurement scheme [16]. Equation (27), not trace preservation alone, is the operational statement that the order of spacelike instruments is gauge at the level of ensembles. V2 packages a Bell trial as one *ActualizationBatch*: the wing instruments remain localized, commuting, and order independent, while the preferred-tick selector consumes the complete joint context and produces the joint record. For every admissible batch its local marginals must be independent of the remote setting.

Order-independence is not selector covariance. It is tempting, and wrong, to conclude that a covariant weight layer delivers a covariant selector. Let μ_f be the measure over complete realized histories induced by a selector law under foliation f, and let $\tau_{f\to g}$ exchange only the birth order of spacelike-incomparable events. The two genuinely distinct further conditions are

$$\mu_g = (\tau_{f\to g})_*\mu_f \qquad \text{and} \qquad (\pi_\mathcal{E})_*\mu_{f_0} = P_\mathcal{E}^{\text{cov}} \text{ for every experiment } \mathcal{E}, \tag{28}$$

the first requiring genuine equivariance and the second requiring only that a physically fixed preferred foliation f_0 be operationally hidden. **Neither follows from** (27), which contains no selector law and no history measure. Keeping these apart is, in our view, the single most common conflation in discussions of relativistic collapse.

9.2 Four representative responses, each priced

The available responses are families rather than an exhaustive classification, and they are not mutually exclusive.

No selector. Deny Postulate 1: unitary theory only, and the covariance problem disappears because there is nothing to make covariant [9]. The price is symmetric and real — the Born weights must then be derived without a selector, whether by decision-theoretic argument [10, 11], contested at length by Kent [13], or envariance [12].

Hypersurface-relative conditionalization. Take the state update to be conditionalization and let it be hypersurface-relative, as in [18, 19]. This is update semantics; it does not by itself say which event becomes actual or with what measure.

Covariant stochastic flashes. Take the actualization events themselves as the primitive ontology [21], for which relativistic models exist for fixed particle number, originally without interaction [22] and subsequently with interaction [23]. Indistinguishable particles, variable number, and full interacting field theory remain open.

A real preferred foliation. Let the selector act on a real but hidden foliation. This makes a total actualization order well defined, and it is where the discrete model of this paper sits, since a global tick order is one of its axioms. The empirically required price is operational hiding — the second condition in (28) — plus ordinary Lorentz recovery in every observable sector. Strong history equivariance, the first condition, is a separate ontological demand and need not be accepted by a genuinely preferred-foliation theory. Section 9.3 conditionally reduces the hiding obligation; Lorentz recovery remains only partial.

9.3 Reducing the hiding debt, and locating the Lorentz debt

The two empirical checks owed by a preferred-foliation ontology are not symmetric. Strong history equivariance is tracked separately.

Proposition 4 (Operational hiding is not an independent obligation)**.** *Let a selector on a fixed foliation f_0 generate a measure μ_{f_0} over complete histories. If* (i) *the marginal of μ_{f_0} on recorded data equals the quantum composed-history weight $P(h) = \mathrm{Tr}[\mathcal{I}^{M_N}_{o_N} \circ \cdots \circ \mathcal{I}^{M_1}_{o_1}(\rho_0)]$ for every experiment, and* (ii) *those weights satisfy* (27)*, then $(\pi_{\mathcal{E}})_* \mu_{f_0} = P^{\mathrm{cov}}_{\mathcal{E}}$ for every operational experiment: the foliation is operationally hidden.*

Proof. By (i) the recorded distribution is $P(h)$. By (ii) $P(h)$ is unchanged when spacelike-separated instruments are exchanged in composition order, and two admissible foliations of one experiment differ exactly by such exchanges; so $P(h)$ is foliation-independent and equals $P^{\mathrm{cov}}_{\mathcal{E}}$. □

The proof is nearly immediate once the hypotheses are stated, and its value is relocation rather than resolution: hiding a preferred foliation requires nothing beyond reproducing quantum statistics. It is one-way — foliation-independent statistics need not be quantum — and it does not deliver the stronger equivariance of (28)'s first equation, which concerns unobservable history detail and is presumably false for a genuinely preferred foliation. Strong equivariance is not required for empirical adequacy; operational hiding is.

The corollary is more informative than the proposition.

Proposition 5 (The barrier is not relativistic)**.** *If a substrate's four CHSH observables are functions on one sample space under a single setting-independent measure, their pushforward is a joint distribution and Fine's theorem gives $|S| \leq 2$. Hypothesis* (i) *of Proposition 4 then fails on Bell experiments, and with it operational hiding. By Proposition 3, making the outcome update noninvertible or energy-dissipating does not alter this conclusion.*

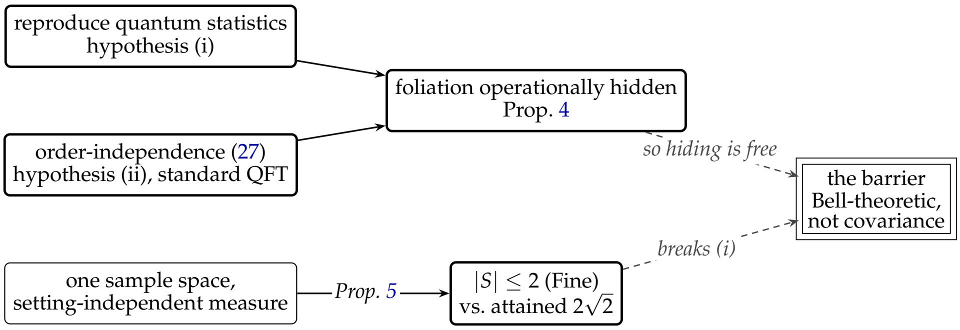

Figure 15: The two reductions in one picture. Hiding a preferred foliation follows from reproducing quantum statistics together with standard order-independence — it is not an independent obligation. What blocks a locally factorized substrate is the lower path: a single setting-independent joint assignment caps CHSH at 2 against the attained $2\sqrt{2}$, which breaks hypothesis (i) directly. V2 preserves measurement independence but rejects that joint assignment through a globally contextual selector. The open obligation (double-ruled) is physical recovery of that response law, which would exist even in a non-relativistic world.

Exhaustive enumeration gives $\max|S| = 2$ over all sixteen deterministic joint assignments against the attained quantum $2\sqrt{2} = 2.828427$, while the order-independence half of Proposition 4 checks numerically to 2.2×10^{-16} over random states and settings. So: *if a substrate reproduced quantum statistics, its preferred foliation would automatically be hidden.* The v2 reference selector evades the corollary's common-response premise by using a complete-context, nonfactorizable map while retaining a setting-independent equilibrium measure. That is a consistency witness only. The physical difficulty is recovering the same statistics from substrate dynamics without target-coded weights, and that obstacle is Bell-theoretic rather than relativistic — it would be present in a non-relativistic world. Figure 15 draws the two propositions as one diagram: what buys hiding, and where the one blocked path actually is.

The second debt is real, and its free sector can be paid exactly. The model's Laplacian is the eighteen-point operator $\frac{1}{3}\sum_{\text{face}} + \frac{1}{6}\sum_{\text{edge}} - 4$, whose symbol expands (in exact arithmetic) to

$$L(\mathbf{k}) = -k^2 + \tfrac{1}{12}\left(k^2\right)^2 - \tfrac{1}{360}\left(\sum_i k_i^6 + 5\sum_{i\neq j} k_i^4 k_j^2\right) + O(k^8). \tag{29}$$

The fourth-order term is a perfect function of k^2: **the stencil is isotropic through fourth order**, and structurally so — the naive seven-point operator has fourth-order coefficient $\frac{1}{12}\sum_i k_i^4$, which is not, and the twelve edge terms are exactly what cancels it. The leading anisotropy is therefore pushed to sixth order, giving a fractional direction-dependent phase velocity

$$\left|\frac{\Delta v}{v}\right|_{\langle 100\rangle - \langle 111\rangle} = \frac{(ka)^4}{3240} + O\big((ka)^6\big). \tag{30}$$

If one inserts the calibration $a = \ell_P$ and treats a photon-scale momentum as the free scalar's k, the highest observed gamma-ray scale gives the formal value 1.4×10^{-56}, far below optical-cavity isotropy bounds near 10^{-18}. This is a conditional parametric momentum-scale insertion, not a photon prediction: no photon carrier or cosmic-ray bound-state pole is constructed here. It serves only to calibrate the tree-level stencil term.

Radiative stability: the tree-level result and the unpaid loop problem. Equation (30) is a tree-level statement, and a small tree-level number is not by itself evidence of Lorentz recovery. The standard objection is that a Planck-suppressed dimension-six violation, inserted in a loop whose integral is dominated by the cutoff, generates a *dimension-four* violation with coefficient of order α/π, requiring tuning of order 10^{-20} against observation [24]. The spatial stencil does contain one useful exact result, but it does not close that objection.

For a general cubic-symmetric eighteen-point operator with face weight w_f and edge weight w_e, the continuum limit requires $w_f + 4w_e = 1$ and isotropy of the k^4 term requires $w_f = 2w_e$; together these have the unique solution $w_f = \frac{1}{3}$, $w_e = \frac{1}{6}$ — the production weights. There is no free parameter in this *tree-level spatial improvement* to be tuned. Averaging a scalar spatial rank-two form $c_{ij}k_ik_j$ over the cubic group leaves $c_{ij} \propto \delta_{ij}$.

That statement is much narrower than dimension-four Lorentz protection. Cubic symmetry does not relate temporal and spatial kinetic coefficients, electric and magnetic terms, or independently renormalized sectors, and it permits native-vector gradients such as $\sum_i(\partial_i J_i)^2$. For a single propagating cone with $\omega^2 = c_0^2(1+\epsilon)k^2$, the rescaling $t \to t/\sqrt{1+\epsilon}$ removes ϵ completely: that sector's speed *is* the definition of the speed, and there is nothing to compare it against. With two sectors the same rescaling leaves $(1+\epsilon_2)/(1+\epsilon_1)$, so only the difference survives. This is one unit redundancy, not a symmetry protection mechanism.

Proposition 6 (Tree-level scalar spatial reduction)**.** *Within the declared scalar eighteen-point spatial stencil, the production weights are the unique face/edge weights giving the normalized continuum term and isotropy through $O(k^4)$. A global time rescaling can normalize one cone. Neither fact constrains all symmetry-allowed dimension-four operators or the relative cones of interacting sectors.*

In particular, saying that a dimension-six operator merely renormalizes itself misses the dangerous channel: loops can feed a symmetry-allowed dimension-four operator. The project's frozen QED-like branch does not find its bare common cone to be a one-loop fixed point without a dimension-four counterterm. That off-shell coefficient is scheme-dependent and is not yet an on-shell observable, so it is evidence against automatic protection rather than a completed phenomenological calculation.

Status: open — hard. The exact result is a tree-level spatial symbol identity. Radiative closure requires an inventory of every symmetry-allowed low-dimension operator for one frozen interacting action, followed by on-shell matching of the full two-point mixing matrix. After one time normalization is removed, every remaining relative-cone eigenvalue must be tested. Any required counterterm is imposed, not emergent. Physical Lorentz invariance is not established here.

9.4 Where the remaining obligation lives: one body or two

The preceding results separate two obligations that are often bundled. Hiding a foliation reduces conditionally to reproducing the quantum history statistics, whose barrier is Bell-theoretic. A common choice of units removes one overall cone normalization, but radiative stability does *not* reduce away: relative temporal/spatial and inter-sector coefficients remain physical. The useful organizing distinction below is operational rather than a closure theorem.

Rotations form the little group of a timelike vector: acting on a body's rest four-velocity $u = (c, \mathbf{0})$, a spatial rotation leaves it fixed, whereas a boost maps that rest frame to a different one. They are not independent group factors — commutators of boosts generate rotations. There is nevertheless a useful operational contrast. An orientation test can reorient one apparatus and compare it against itself; a comparison of standards in distinct rest frames needs clocks and rods that instantiate those frames. The table is a heuristic for constructed-standard tests, not an exhaustive partition, and boost-sensitive one-body dispersion tests are an explicit overlap:

	single-apparatus orientation	**cross-frame standards**
what is compared	apparatus with itself, reoriented	one frame against another
typical test	isotropy, sidereal variation	time dilation, cone equality
status here	scalar spatial stencil improved at tree level	**open**

The tree-level spatial-symbol result was a statement about the point group O_h — a rotation group — and the residual (30) is a one-body observable: reorient the apparatus and look for a sidereal signal. The broader dimension-four and radiative questions remain open. The cross-frame standard statements are open as well: whether the matter and field cones agree, whether interacting vertices preserve a common cone, and whether composite clocks and rods transform correctly. The single-cone absorption argument of Section 9.3 says the same thing from the other side: one cone can always be scaled to unity, while a *ratio* of two cones is invariant under every global rescaling and cannot be defined away.

Operational convention [SCOPE]. Tests that compare rates, worldlines, or transported standards between frames require physical standards sufficient to instantiate that comparison. This programme therefore asks its substrate to construct a clock before claiming an operational time-dilation test. This is a methodological requirement, not a theorem that every two-frame observable is mathematically undefined without two clock parametrizations: geodesic deviation, for example, uses one reference proper time and a Jacobi field.

The scope word is load-bearing and an earlier draft omitted it. Not every boost-*sensitive* quantity needs a constructed clock: the mass-shell diagnostic of Section 9.5.1 tests a free dispersion's Lorentz form with no constructed clock, and it is a one-body calculation. What remains clock-gated in this programme is the claim that an actual substrate-built clock obeys the target dilation law.

This scope distinction reorganizes the recovery contract rather than discharging it. The one-body dispersion register is computed, while interacting matter, clocks, and rods remain open. The clock construction Section 4 priced is one concrete missing standard, not a universal obstruction to defining every relativistic observable. It is also needed if the stage index is to be read as elapsed time: Section 3 argued that something physical must carry that index, and Section 10 returns to the obligation.

The same restraint applies past special relativity. Operational curvature requires physical free-fall systems and standards, but geodesic deviation itself may be parametrized by one reference proper time and a separation field; it does not prove a universal two-clock gate.

One component of the two-body register can be sharpened immediately. The model's wave speed $1/\sqrt{3}$ is not the stability bound: for the eighteen-point operator the symbol attains $\max|L| = 16/3$ on the face diagonal, giving $C \leq \sqrt{3}/2$. It is instead exactly the largest isotropic speed whose round effective cone is *contained* in the octahedral causal polytope, whose facets $\sum_i \pm x_i = 1$ lie at distance $1/\sqrt{D}$ from the origin. That is the same $1/\sqrt{D}$ as the per-axis component of an isotropic unit spread, since $\mathbb{E}[n_i^2] = 1/D$ for a uniformly random unit vector — one isotropic unit distributed over D orthogonal axes, counted two ways. The characterization is exact, and it relocates the choice rather than removing it: the containment bound is $1/\sqrt{3}$ only for six-neighbour causality, whereas the eighteen-point Laplacian's reach and the model's stated Moore locality both give 1. The cone speed is therefore an interior selection whose adoption carries a commitment about which neighbourhood defines causal reach.

The neighbourhood is a model definition, but its consequences are exactly distinguishable. For the stated eighteen-point wave update, a point perturbation produces first-tick face amplitudes 1/9, edge amplitudes 1/18, and zero corner amplitudes. Six-, eighteen-, and twenty-six-neighbour support therefore disagree already at $t = 1$, and exact dependency can be tracked symbolically at any amplitude. Machine epsilon is not a support criterion: it is the relative spacing near unity, not the smallest representable nonzero number.

What the measurements do show is an *effective* separation of scales. In the reported point-source run, the amplitude at the M18 support edge falls by 1.19 decades per tick — 2.5×10^{-7} of the peak at $t = 8$, 2.7×10^{-17} at $t = 16$, and 2.0×10^{-26} at $t = 24$. That finite-run suppression can make the precursor irrelevant to an instrument with a declared dynamic range; it does not make strict support definitional or unobservable in principle.

The strict M18 support is a cuboctahedron, anisotropic by 34.3% between its face and edge directions. Threshold contours of the measured Green function are much rounder: 0.71% anisotropy near the effective cone and 7.2% even fifteen decades down in that run. Their residual anisotropy decreases empirically as $t^{-1.1}$ to $t^{-1.3}$ depending on probe depth, consistent with an Airy-front expectation of $t^{-4/3}$. These are measured effective-field profiles, not the exact causal boundary and not a proof of a uniform continuum-limit rate.

None of this closes anything. It is a one-body statement — an independent cross-check of the free-sector result above by a different method — and the obligation that matters is the two-body one.

9.5 The two-body bill, itemized

Six line-items follow. The free massive-mode expansion and the free multiparticle threshold-cone shift are computed at their stated scope. An exact symbol decomposition shows how a free cross-sector cone could be matched, but no working matter sector is built. The static-well substitution is exploratory and inconclusive under the corrected finite-gap comparator, while the soliton shape-mode campaign is currently unreproduced and fails an independent common-cone check. Each item names what survives and what remains open.

9.5.1 Free massive modes

That obligation, however, was bundled too coarsely, and unbundling it turns part of it into a calculation.

Begin with what will *not* work. The natural test is to take the clock of Section 4, give it momentum, and compare its period to γ. It is vacuous, for the reason recorded when that clock was bought: a mechanical framework is Galilean invariant and reports zero dilation whatever the substrate does. Dilation in a field substrate comes from the binding being mediated at the substrate's own speed — Lorentz's electron-theory reasoning, and the light-clock argument in modern dress — so the object to interrogate is the *dispersion of a massive mode*, which needs no clock at all.

For a free mode, a packet around wavenumber k moves at group velocity $v_g = |\nabla_k \Omega|$. Relativistic mass-shell dispersion implies the diagnostic

$$\Omega_{\text{shell}}(k) = \Omega(k)\sqrt{1 - v_g(k)^2/c^2} = M, \tag{31}$$

an identity whenever $\Omega^2 = c^2k^2 + M^2$. This is a mass-shell diagnostic, not a constructed internal-clock observable: constancy tests the free dispersion's Lorentz form but does not by itself demonstrate that a physical clock made from the species dilates. Adding a mass to the leapfrog gives the exact dispersion $4\sin^2(\Omega/2) = C^2(-L(k)) + M^2$, and expanding along [100], [110] and [111] returns *identical* coefficients through k^4:

$$\Omega^2 = \left(M^2 + \tfrac{M^4}{12}\right) + \left(\tfrac{1}{3} + \tfrac{M^2}{18} + \tfrac{M^4}{90}\right)k^2 - \left(\tfrac{1}{54} + \tfrac{M^2}{1080}\right)k^4 + \dots$$

Reading the k^2 coefficient as the squared limiting speed,

$$\frac{C_{\text{eff}}(M)}{C} = 1 + \frac{M^2}{12} + \frac{19M^4}{1440} + O(M^6). \tag{32}$$

The limiting speed is thus *exactly isotropic* — the three symmetry directions agree identically, not approximately — and *depends on the rest mass*. The second is a negative result: two species of different mass do not share a cone exactly. It is the classic species-dependent maximum-attainable-velocity signature.

Within a species the free dispersion is nearly Lorentzian. Using that species' own C_{eff}, the residual in (31) is $O(k^4)$ and agrees with $k^4/(36M^2)$ to within 1.1% over the sampled masses and momenta; in the physically natural variable $\beta = v_g/C$ this is

$$\frac{\Delta\Omega_{\text{shell}}}{\Omega_{\text{shell}}} \simeq \frac{M^2\beta^4}{4}. \tag{33}$$

Each mass therefore carries its own very nearly Lorentzian free sector, and the violation lives in the mismatch *between* masses in this scalar free sector. If the paper's tick-to-Planck mapping is inserted, electron- and proton-sized mass parameters give a formal difference 1.6×10^{-40} and an electron-sized mode at $\beta = 0.1$ gives a residual 1.5×10^{-50}. These are parametric evaluations of a free-mode mass-shell diagnostic, not predictions for physical electrons, protons, or clocks; the required matter sectors and interacting matching are not built.

The same expansion reproduces the free-sector anisotropy of Section 9.3 independently, giving $(v_{[100]} - v_{[111]})/C = -k^4/3240$ exactly, which validates the chain.

One scope limit matters more than the result: this concerns *one sector with a varying mass*, whereas the far larger problem is that distinct sectors of the model carry leading speeds differing at order unity. That problem is the subject of the next subsection, and it turns out to be the one place in this paper where a stubbornly negative result inverted.

9.5.2 A common cone across sectors

One sector can always be given unit speed by choosing the time unit. That removes one normalization and no more: if two propagating sectors have slopes c_A and c_B, the ratio c_A/c_B survives every common unit change and is physical preferred-frame data. The discrete model carries $c^2 = 1/3$ in its production field sector and raw 1 in a standalone Wilson matter sector — an order-unity mismatch, against which the 10^{-40} of the previous subsection is irrelevant. This is the real obligation.

The first three findings are negative, and sharply so. Writing $u_i = \cos q_i$, matching a scalar Wilson parameter r to the field symbol identically, $c_s^2[3 - \sum u_i^2 + r^2(3 - \sum u_i)^2] = \kappa\,(-L_{18})$, gives ten coefficient equations of which two suffice: the u_1u_2 coefficient forces $\kappa = -3c_s^2r^2$ and the u_1 coefficient forces $\kappa = +9c_s^2r^2$, so $c_s^2r^2 = 0$. *No nontrivial real (c_s, r) gives an exact all-momentum match.* Second, the O_h-invariant polynomials are generated by $\sum q_i^2$, $\sum_{i<j} q_i^2 q_j^2$ and $q_1^2q_2^2q_3^2$, so the number of isotropy conditions grows without bound — cumulatively $0, 1, 3, 6, 10, 16, 23, 32$ through order q^{16}. Under naive order-by-order finite-shell tuning, this is an unbounded sequence of conditions. The counting alone is not a no-go against a finite all-orders identity: one structure may satisfy infinitely many Taylor conditions at once, as the seven-square identity below demonstrates. Third, and least expected, enriching both structures with a face-diagonal shell yields six equations in seven Gram monomials, which *solve*; what fails is realizability. For a real kinetic vector the Gram $A_{st} = a_s a_t$ is an outer product and must have $A_{11}A_{22} - A_{12}^2 = 0$, whereas the solution forces $A_{11}A_{22} - A_{12}^2 = \frac{2}{9}\kappa^2$. The required Gram has rank two; the obstruction is structural, not arithmetic, and being proportional to κ^2 it survives every rescaling. Adding the fifth Clifford structure available to four-component spinors leaves the residual unchanged; adding a third kinetic shell makes the rank conditions insoluble.

An economical constructive resolution uses a half-offset carrier basis. Regrouping and applying $1 - \cos A = 2\sin^2\frac{A}{2}$ and $1 - \cos A \cos B = \sin^2\frac{A-B}{2} + \sin^2\frac{A+B}{2}$ exhibits the field symbol as a *sum of squares*, and every argument that appears is a half-angle.

This factorization does *not* make half-offset sites necessary. For any integer hop d with graph weight w,

$$2w[1-\cos(q\cdot d)] = w\sin^2(q\cdot d) + w[1-\cos(q\cdot d)]^2. \tag{34}$$

Applying (34) to the three face-pair hops with $w = 1/3$ and six edge-pair hops with $w = 1/6$ gives an exact *eighteen*-square integer-frequency decomposition of $-L_{18}$. Equivalently, the phase $e^{iq/2}$ in $e^{iq} - 1 = 2ie^{iq/2}\sin(q/2)$ shows why a half-angle magnitude alone does not locate endpoints at half sites. Under the termwise Clifford ansatz below, naively linearizing the eighteen integer-hop squares uses spinor dimension 512. The half-angle construction reduces that ansatz's known exact count to seven, at dimension eight. It is an economy result, not a theorem that the physical lattice must be refined. If this cheaper basis is physically instantiated, whether it is read as refined sites or staggered cochains is a separate, open choice.

Under the declared termwise Clifford linearization $H = \sum_\mu \Gamma^\mu \phi_\mu$, imposing cancellation of every cross term separately makes the Γ^μ mutually anticommute, and dimension 2^k carries $2k+1$ such structures. This is a construction price, not a lower bound on every matrix factorization: algebraic relations among the ϕ_μ could permit collective cross-term cancellation. Within this declared ansatz, the first decomposition we found was not the shortest, and the difference matters by a factor of four. Restricting to nearest-neighbour hops on the half-offset lattice — degree at most one per variable in $x = q/2$ — and writing $s_i = \sin x_i$, $c_i = \cos x_i$, there is an exact and cubic-covariant decomposition into *seven*:

$$-L_{18} \;=\; 4\sum_i s_i^2 c_j^2 c_k^2 \;+\; \tfrac{16}{3}\sum_i s_j^2 s_k^2 c_i^2 \;+\; 4\, s_1^2 s_2^2 s_3^2, \tag{35}$$

with $\{i,j,k\} = \{1,2,3\}$. It is checkable by hand: with $A = s_1^2$ etc., $S_1 = A+B+C$ and $S_2 = AB+BC+CA$, the three groups contribute $4[S_1 - 2S_2 + 3ABC]$, $\frac{16}{3}[S_2 - 3ABC]$ and $4ABC$; the triple products cancel exactly, $12 - 16 + 4 = 0$, leaving $4S_1 - \frac{8}{3}S_2 = \frac{4}{3}[3S_1 - 2S_2] = -L_{18}$. The lone singlet is not decoration — it is what kills ABC. A search over the same class finds a non-covariant four-square numerical candidate with fresh-momentum residual 5.3×10^{-15} and does not find three. No symbolic or interval existence certificate is supplied for the four. The Hessian rank supplies the rigorous floor $n \geq 3$, while the exact seven below supplies the proven upper bound, so $3 \leq n_{\min} \leq 7$; "none found at three" and the four-square candidate are numerical evidence, not proofs. An exact cubic-covariant seven-square construction gives the upper bound $n_{\text{cov}} \leq 7$; mixed-sector covariant constructions with five or six squares have not been excluded. In the termwise Clifford ansatz, this seven-square construction uses spinor dimension eight; the numerical non-covariant four candidate would use four if certified. *Repository correction.* An older FTD-0798 ledger sentence asserted a local cubic-covariant three-square decomposition, but supplied neither the three factors nor a reproducing artifact; a repository-wide search found no such certificate. That sentence is withdrawn as unsupported. It does not attain the rigorous floor used here; a recovered exact identity would have to be published and checked before changing these bounds. (An earlier draft of this passage reported the minimum as five; its search drew four hundred starts from a single initialization scale, which samples one basin many times rather than many basins once. Dispersing the scale finds the four within tens of starts. The correction is retained here because the method failure is as instructive as the number.) Figure 16 draws the mismatch, its cause, one decomposition, and the price.

If the economical half-angle basis is physically instantiated as hops, which placement classes appear decides what it costs the primitive ontology. Three half-offset shells are available — face $(\pm\frac{1}{2},0,0)$, edge $(\pm\frac{1}{2},\pm\frac{1}{2},0)$ and body centre $(\pm\frac{1}{2},\pm\frac{1}{2},\pm\frac{1}{2})$ — and the restriction now cuts finely: confined to body centres alone the search reaches five and stalls there over two thousand dispersed starts, while the best-found numerical four uses the face and edge shells. Thus the search suggests that leaving the body centres buys one square, but neither search establishes a minimum. There are two priced readings. On a refined-site reading, the half offsets are physical sites of a cubic lattice with spacing $a/2$; that amends the discrete-space postulate and requires recalibrating a_{phys} and every dimensional prediction that depends on it. On a staggered-cochain reading, the Bravais spacing remains a but the theory gains explicit carrier types on cell faces, edges and bodies. Because the numerical four-square candidate uses face and edge shells, that reading requires corresponding staggered 1- and 2-cochain content, not merely body-centred sites. Neither reading is adopted here. Only the cochain reading would leave a_{phys} unchanged. (The stall at four is search evidence of the same kind that previously mis-reported the minimum, and is offered with exactly that confidence; the rank bound $n \geq 3$ is the only proof in this paragraph.)

The seven-square construction is moreover not a numerical search outcome. The eight half-angle triple products $\prod_i g_i(q_i/2)$ with $g_i \in \{\cos, \sin\}$ satisfy $\sum(\prod_i g_i)^2 = \prod_i(\cos^2 + \sin^2) = 1$ identically. One of the eight is the pure cosine $S = \prod_i \cos(q_i/2)$ — which is exactly the body-centred-cubic nearest-neighbour structure factor, the eight neighbours sitting at $(\pm\frac{1}{2}, \pm\frac{1}{2}, \pm\frac{1}{2})$ — and the remaining seven have equal-weight sum $1 - S^2 = (1 - S)(1 + S)$. Equation (35) is diagonal in that seven-element half-angle basis, which is why it is spatially cubic-covariant and why it has seven squares. The factorization aligns it algebraically with the BCC complement $1 - S$, but does not put the usual BCC Laplacian $8(1 - S)$ in the same constant-weight square cone: the extra factor $1 + S$ is load-bearing.

Two consequences deserve stating. Structurally, the optional half-angle basis aligns algebraically with the body-centred structure factor that also appears in a Watson-type period identity and an SU(3) factor. This is a shared representation pattern, not a derivation that the physical site lattice must be body-centred. Within the termwise Clifford ansatz, a mass structure is chosen to anticommute with every kinetic structure, so it costs one further structure. The numerical four-square candidate would leave one structure spare and would admit a mass at dimension four *if certified*; no rigorous low-rank mass price follows yet. The exact cubic-covariant seven-square construction saturates dimension eight under that ansatz, so adding a termwise-anticommuting mass to *that construction* uses dimension sixteen. A lower-rank mixed-sector cubic-covariant construction could reduce the price and has not been excluded. (An earlier draft, built on the mis-reported five, concluded the opposite — that the minimal carrier was necessarily massless; that prediction died with the five and is recorded here so it cannot be cited from memory.)

What this does and does not close. At the scalar, spatial, tree-level symbol scope, it shows that exact slope matching is not algebraically obstructed: there is an explicit decomposition, and its termwise-Clifford construction price is stated. It does not deliver a working matter sector. Whether the resulting operator is an acceptable fermion — doubling, chirality, and the locality of the induced interactions — is untested; the time discretization is a separate matching problem that can only add conditions; a spinor's worth of anticommuting structures is not a ternary state, so the carrier must either *emerge* from the primitive states or be *adopted* as a new type at a price; and the order-unity mismatch still stands in the model as it is actually run. Interacting vertices remain open.

9.5.3 What a composite inherits

The calculation performed here is exact at its stated order but answers a narrower question than an earlier draft claimed. Let free constituents have $\omega_a(k) = m_a + k^2/(2\mu_a) + O(k^4)$. Minimizing their summed energy at fixed total momentum gives the bottom of the free multiparticle continuum, not a bound-state pole:

$$E_{\text{thr}}(K) = \min_{\sum_a k_a = K} \sum_a \omega_a(k_a) = \sum_a m_a + \frac{K^2}{2\sum_a \mu_a} + O(K^4). \tag{36}$$

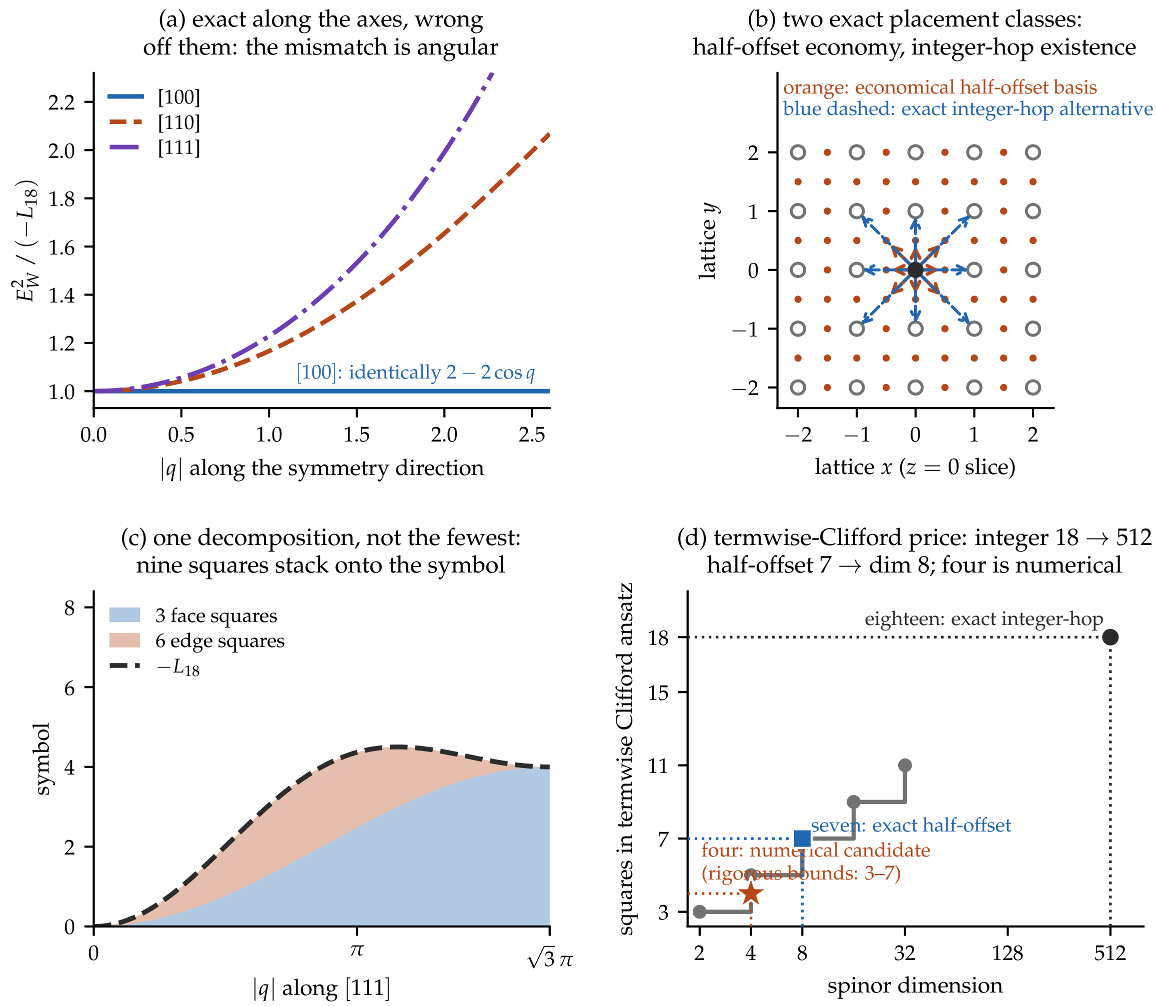

Figure 16: Why two sectors miss, and what an exact hit can cost. (a) Normalised to a common slope, the flux symbol and a Wilson operator agree *identically* along the cubic axes — both are $2 - 2\cos q$ — and diverge only off them, reaching 1.65 along [110] and 1.99 along [111] at $|q| = 2$. The mismatch is purely angular, so no rescaling is even the right shape. (b) Two constructive placement classes in the $z = 0$ slice: the economical half-angle basis uses half-unit face/edge placements (orange), while an exact but longer integer-frequency construction uses unit face/edge hops (blue dashed). Half offsets reduce the known price; they are not forced by existence. (c) One half-angle identity: three face squares and six edge squares stack exactly onto $-L_{18}$, with residual 1.8×10^{-15} over 4000 random momenta; Equation (34) independently gives an exact eighteen-square integer-hop identity. (d) The known exact and searched termwise-Clifford prices. Naive Cliffordization sends the integer-hop eighteen to dimension 512. An exact spatially cubic-covariant *seven*-square identity holds, $-L_{18} = 4\sum_i \sin^2 x_i \cos^2 x_j \cos^2 x_k + \frac{16}{3}\sum_i \sin^2 x_j \sin^2 x_k \cos^2 x_i + 4\sin^2 x_1 \sin^2 x_2 \sin^2 x_3$ with $x = q/2$, using dimension 8 in that ansatz; and a search over the half-offset nearest-neighbour class finds a non-covariant numerical *four*, which would fit at dimension 4 with one termwise-anticommuting structure spare for a mass if certified. The rigorous unrestricted floor is three, so the rigorous unrestricted bounds remain [3, 7]; four is a numerical candidate without an existence certificate, and seven is an exact spatially cubic-covariant construction rather than a proven cubic-covariant minimum.

Writing $\delta_a = (c_a^2 - C^2)/C^2$ and expanding to first order gives the corresponding free-threshold shift

$$\delta_{\text{thr}} = \sum_a w_a \delta_a, \qquad w_a = \frac{\mu_a}{\sum_b \mu_b}, \quad \mu_a = \frac{m_a}{c_a^2}. \tag{37}$$

The exact-dispersion calculation checks this free result: unequal pairs match (37) to eight significant figures, and the normalized threshold-cone shift $E_{\text{thr}}(0)E''_{\text{thr}}(0)/C^2 - 1$ for N identical free constituents is constant to seven digits for $N = 1$ through 30. The N-independence is therefore a free-threshold identity, not evidence that a bound macroscopic body inherits the same cone.

A bound composite instead has an interacting eigenvalue $E_B(K)$, with low-momentum cone determined by $E_B(0)E''_B(0)$. Binding fields, stresses, and interaction-dependent boost generators contribute to both factors. Even a momentum-independent binding shift changes $E_B(0)$ while leaving the free curvature in (36) unchanged. Consequently the nucleon, kilogram-body, and Earth extrapolations in an earlier draft were unsupported and are withdrawn. Interacting composite inheritance and every macroscopic common-cone claim remain **open**. Figure 17 now labels the surviving calculation as a free multiparticle threshold.

9.5.4 What a potential is, and where the gate actually lies

It is worth being exact about why the mechanical clock fails, because the obvious diagnosis is the wrong one and it mislocates the remaining work.

A potential $V(|q_i - q_j|)$ is not a primitive of a discrete substrate. A single cell is a state and a local update rule; it has neither a distance nor a potential, both being emergent *two-body* descriptions obtained by localizing two sources, integrating out the field between them, and letting the propagation time vanish. Two independent limits are taken in that sentence, controlled by different parameters: the constituents' dispersion is replaced by $p^2/2m$ (controlled by v/c), and the retarded exchange is replaced by an instantaneous $V(r)$ (controlled by $\omega r/c$).

These two control parameters diagnose separate approximations, but the free-threshold calculation does not prove that only the first matters for a bound state. In covariant QED an instantaneous Coulomb-gauge term is one part of a larger interacting theory; field momentum and interaction terms also enter the boost generator. Hydrogen therefore cannot be used here as proof that binding supplies only weights.

The mechanical clock of Section 4 takes both limits, and its own parameters license neither. At $A = 0.30$ its nodes move at $v_{\max}/C = 0.156$, so the Newtonian substitution discards $(v/C)^2 = 2.4\%$; and $\omega r/C = 0.62$ for unit spacing, rising to 1.87 across the chain, against 3.6×10^{-3} for an atom and 0.30 for a heavy nucleus. It is more relativistic internally than a nucleus.

So the clock is not the wrong *kind* of object; it is under-modelled. Replacing its nodes' dispersion is a legitimate fixed-model test, but the test must be judged against the finite-gap relativistic comparator rather than an infinitesimal-clock approximation.

9.5.5 The substitution, executed

Two constituents on the axial lattice, bound by a finite square well in the relative coordinate, give an exactly diagonalizable relative-motion problem at each total momentum K. Two bound states below the two-particle continuum supply a genuine internal clock, of gap $\Omega = E_1 - E_0$. Let $M_0 = E_0(0)$ and $M_1 = E_1(0)$. For two exact relativistic mass branches sharing speed c, the finite-gap target is

$$R_{\text{SR}}(K) = \frac{\Omega_{\text{SR}}(K)}{\Omega(0)} = \frac{M_0 + M_1}{\sqrt{M_0^2 + c^2K^2} + \sqrt{M_1^2 + c^2K^2}}. \tag{38}$$

The familiar $1/\gamma_0$ law is only the limit $\Omega(0)/M_0 \to 0$. Consequently, fitting $\Omega(K)/\Omega(0) = \gamma_0^p$ and demanding $p = -1$ at finite gap is not an exact relativity test.

The control behaves as expected. With Newtonian nodes the internal gap is K-independent to 8.7×10^{-6} while γ_0 reaches 1.399: the Galilean model supplies no clock slowing.

The corrected comparison is inconclusive, not a structural negative. Using each measured ground-state γ_0 to eliminate cK in (38), the six fixed cases at $K = 0.2$ give observed-to-SR ratios 0.9892, 0.9936, 0.9963, 1.0000, 0.9934 and 0.9948. Thus one case is indistinguishable at the quoted precision and the others miss by only 0.4–1.1%. The old exponent range $[-2.700, -0.937]$ exaggerates this discrepancy because it compares finite gaps to the infinitesimal-gap $p = -1$ limit. For the free pair define $\gamma_f = \omega(K/2)/\omega(0)$. The relation $\mu_K/\mu_0 \simeq \gamma_f^3$ is only a low-momentum continuum approximation here: its tabulated relative residual reaches 6.5%, so it cannot support an exact mechanism diagnosis.

The static well therefore neither establishes universal dilation nor proves its failure. Finite-volume, discretization and held-out-momentum controls against (38) are still required. More generally, a substrate claim needs constituent dispersion, binding fields, stress-energy and boost generators to arise from one interacting dynamics; this small fixed-potential instrument does not provide that closure.

One further issue is independent. At every amplitude the mechanical clock's internal frequency lies inside the propagating band, $\omega < 2 \arcsin C = 1.2310$, so it lacks kinematic band clearance and can radiate if a nonzero allowed coupling to travelling modes exists; no radiative lifetime is computed. But the composites above are bound *below* the continuum at every K — band clearance holds there by construction — and the corrected dilation test remains inconclusive. Clearing the band therefore does not settle covariance.

9.5.6 One covariant interacting action: the candidate test

The corrected comparison removes the earlier claimed no-go and its mechanism diagnosis. Termwise prescriptions such as contracting a written well radius are exploratory model changes, not covariance proofs. The surviving requirement is that kinetic, binding-field, stress-energy, and boost generators arise from one covariant interacting action rather than from transformation rules attached term by term.

A continuum φ^4 kink is therefore a sensible candidate: one action supplies a soliton and a bound internal shape mode below the continuum. The present lattice artifact, however, does **not** establish time dilation. It starts from an imported continuum-boosted profile, radiates, and gives $p \simeq -1.14$ to -1.18 when the independently declared substrate speed is used. The Lorentz square-root form reaches $p \simeq -1$ only after fitting a carrier-specific speed about 6% below the speed inferred from the kink energy. A covariant carrier must have one common cone. More seriously, the committed runner has a different parameter set from the quoted campaign and does not reproduce these numbers.

> **Status: open — unreproduced and common-cone mismatched.** The static-well comparison is exploratory and inconclusive under the corrected finite-gap comparator. The one-covariant-action requirement is a programme criterion, not a result of that screen. The soliton shape-mode clock is a candidate only. A valid campaign must lock one runner, track the centre velocity, project onto a comoving internal eigenmode, measure radiation and finite-volume convergence, approach the continuum, and test held-out velocities against the independently measured energy cone. Until then no Lorentz-recovery claim is made from the kink.

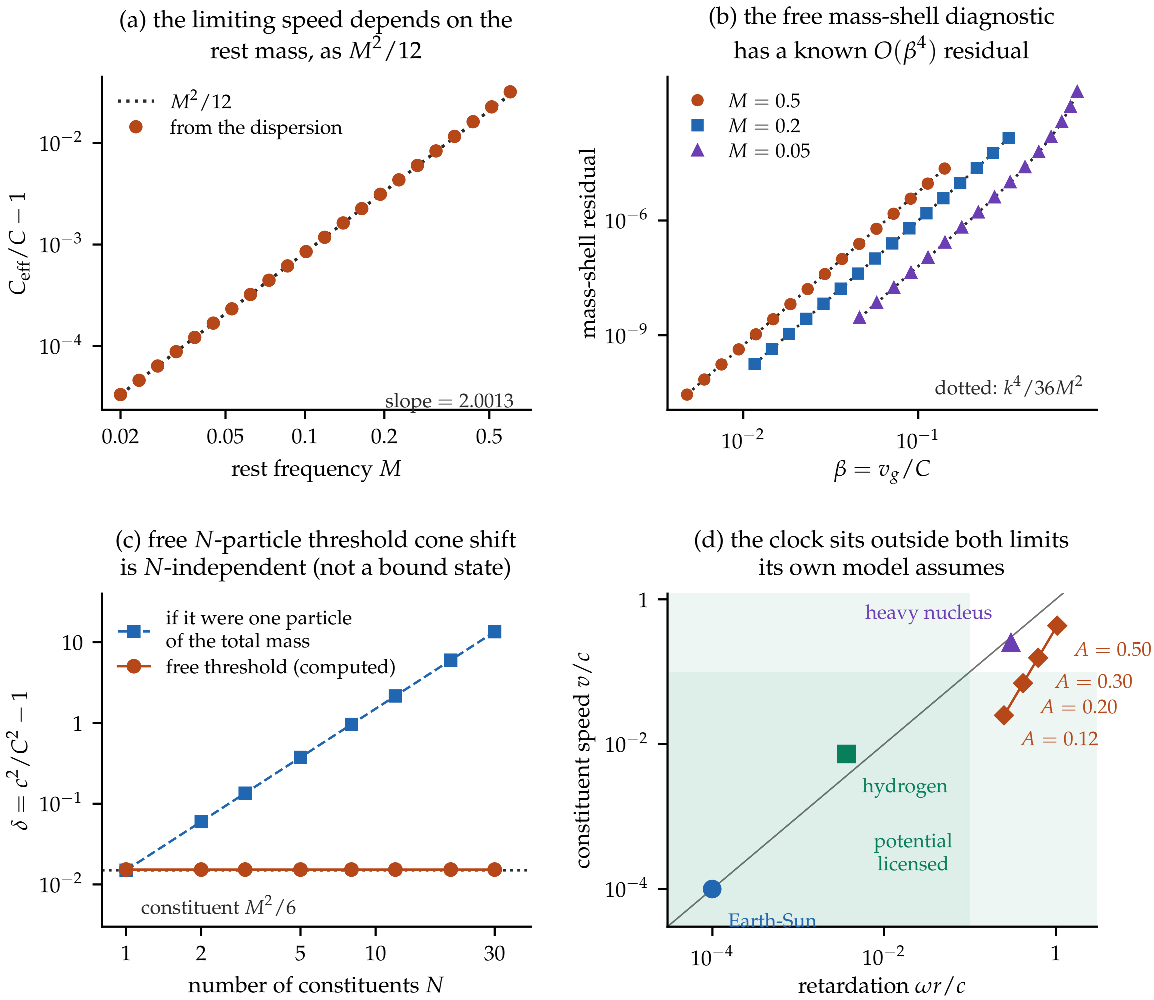

Figure 17: The cone arc of Sections 9.5.1–9.5.4, computed. (a) The limiting speed extracted from the exact dispersion tracks $M^2/12$; the fitted log–log slope is 2.0013. (b) Using each species' own $C_{\rm eff}$, the free mass-shell diagnostic has an $O(\beta^4)$ residual that follows the analytic $k^4/36M^2$ (dotted) to within 1.1% across three masses. (c) The free-threshold result: the normalized cone shift $E_{\rm thr}(0)E''_{\rm thr}(0)/C^2 - 1$ at the bottom of an identical-particle continuum is constant to seven digits as the particle number runs $1 \to 30$, while treating the total mass as one particle grows as N^2. This is not a bound-state result. (d) The two limits hidden in a distance potential, with real systems placed. Orbits lie on the diagonal because $\omega r = v$ for them; the mechanical clock does not, and sits outside the licensed corner on both axes — more relativistic internally than a heavy nucleus.

9.6 The discrete diagnostic

The obligations above were all continuum-shaped: a cone, a boost, a dispersion. A discrete substrate admits one further test that has no continuum analogue, and it is worth recording even though it decides nothing here.

If a theory supplies a coarse-graining from its histories to locally finite causal sets, the recursion (25) becomes accretion of one event at a time onto a partial order, and covariance acquires a discrete form: the probability of reaching a given unlabelled causal set must not depend on the labelled growth path, $\prod_{r\in\gamma} t_r = \prod_{r\in\gamma'} t_r$. Under internal temporality, Markov normalization, discrete general covariance *and Bell causality*, the classical solutions are classified [25]; the quantum case remains an open programme [26, 27]. Two precision points are worth carrying: the birth label of an element is gauge, whereas the stage number $n = |C_n|$ is label-invariant and is not gauge, though neither is it a coordinate time. We record this as an *optional diagnostic*: it is neither necessary nor sufficient for hiding a preferred lattice, and it does not replace the Lorentz-recovery obligation above.

The division leaves a bill: the one-body half closed with numbers, the two-body half open, priced, and gated on an instrument this paper costed in Section 4 and the substrate did not supply. The next section observes that the architecture's own stage index is gated the same way — which is the promise left standing in Section 3.

10 Quantum gravity, and the clock again

Canonical quantization of generally covariant general relativity yields the Hamiltonian constraint and the familiar problem of time, schematically $\hat{H}\Psi = 0$ [28]. The standard relational repair conditions on a clock subsystem [29], recovering evolution as conditional potentiality, with known objections about localization and propagators [30] that are partly addressed by covariant clock observables [31] and complicated again for finite-resource clocks [32].

For this paper the point is structural and closes a loop. The v2 contract keeps three quantities separate:

$$\begin{aligned} n \quad & \text{ontic global update order,} \\ \phi_x(n), \tau_x \quad & \text{local phase and operational duration,} \\ k_x \quad & \text{compliant actualization-gate count.} \end{aligned} \tag{39}$$

The first is postulated and does not need to be read from a clock. The second turns succession into operational duration and must be carried and maintained inside the history. The third counts only fixed-phase crossings that pass the clock-compliance tolerances. For the selected quartic carrier $TA = \sqrt{\pi} G^* \sqrt{m/(2\lambda)}$: G^* fixes the cadence of eligibility, not the global order and not the outcome. **The clock programme is therefore the precondition of local duration and gating, not the source of the world's global update index.**

One further hypothesis class connects gravity to Σ directly, by making the gravitational self-energy difference set an actualization timescale, $\tau \sim \hbar/E_G$ [33, 34]. Its parameter-free version is excluded by underground radiation tests [35], and the Markovian model is further constrained by XENONnT [36]. We note the structural resemblance to Section 8's slow-channel result — a gravitational sector supplying the slow gate — and we label it exactly as what it is: a speculative alignment between two independently derived shapes of requirement, not evidence for either.

That closes the loop the construction opened. The v2 reference pipeline now runs the directed composition $n \to$ substrate history $\to \omega \to$ local instruments $\to G^*$ gate $\to \Sigma \to$ records, with the controller audit carried alongside it. Because its preparation map and

Born weights are adopted, it is a standalone mathematical consistency model, not a physical substrate realization. The older side-by-side check below remains useful only as an independent-module audit.

11 A side-by-side minimal demonstrator

The preceding sections established their results one instrument at a time. A smaller bookkeeping question remains: can their declared modules be evaluated at one parameter set without a transcription contradiction?

This section records substrate, clock, register, gate, and noise calculations in one computable artifact, small enough to run in under a minute. Three identities can be checked to machine precision. It is a side-by-side demonstrator, not an instrument of record: no claim of Sections 4–8 depends on it, and it establishes no composed physics.

The modules in this older artifact are independent computations: no module consumes another's state, and there is no feedback anywhere in it. The test catches shared-constant and transcription inconsistencies; it does not show that the mechanisms coexist in one dynamical system. Separately, the v2 reference simulator composes history, preparation, instruments, maintained gate, selector, and records through explicit interfaces. It still imports the state–effect weights and does not implement physical register-to-carrier backreaction, so a substrate-native coupled model remains open.

11.1 The ideal normal-form clock law, exactly

The carrier's mirror-even mode reduces at leading quartic order to $m\ddot{Q} = -4\lambda Q^3$. Energy conservation gives $\frac{m}{2}\dot{Q}^2 + \lambda Q^4 = \lambda A^4$, so a quarter period is

$$\frac{T}{4} = \frac{1}{A}\sqrt{\frac{m}{2\lambda}}\int_0^1 \frac{du}{\sqrt{1-u^4}} = \frac{1}{A}\sqrt{\frac{m}{2\lambda}}\cdot\frac{\sqrt{\pi}\,G^*}{4},$$

using $\int_0^1 (1-u^4)^{-1/2}\,du = \frac{1}{4}B(\frac{1}{4},\frac{1}{2})$. Hence the ideal normal-form period law

$$T\cdot A = \sqrt{\pi}\,G^*\sqrt{\tfrac{m}{2\lambda}}, \qquad G^* = \Gamma(1/4)/\Gamma(3/4). \tag{40}$$

Panel (b) of Figure 18 shows this without any rescaling: the three trajectories have $A = 0.30, 0.20, 0.12$ and periods $17.5, 26.2, 43.7$. Panel (c) integrates six amplitudes spanning a factor of six at tolerance 10^{-12}, locates each far turning point by event detection, and recovers a fitted log–log slope of -1.000000000 and $\max_A|G^*_{\mathrm{rec}}/G^* - 1| = 1.61\times 10^{-13}$, where G^*_{rec} inverts (40) with no free parameter. The recovery sits at integrator tolerance, which is the correct outcome for an identity rather than a fit.

The significance is the contrast, not the constant. A quadratic well would give π and isochrony; the quartic well gives G^* and amplitude dependence. That is the difference between a clock whose rate is independent of how hard it was struck and one whose rate is not — and it is the reason Section 4 treats the lemniscatic clock as a purchase rather than a gift.

11.2 Testing the one-constant, three-role claim

The architecture's economy claim is that a single energy scale prices three apparently unrelated things: the depth of the well that binds the carrier, the quartic stiffness λ that sets the clock rate through (40), and the Arrhenius barrier in $\tau_{\mathrm{flip}}\nu_0 \sim e^{\varepsilon/T}$. The commitment is that these are not three independent constants.

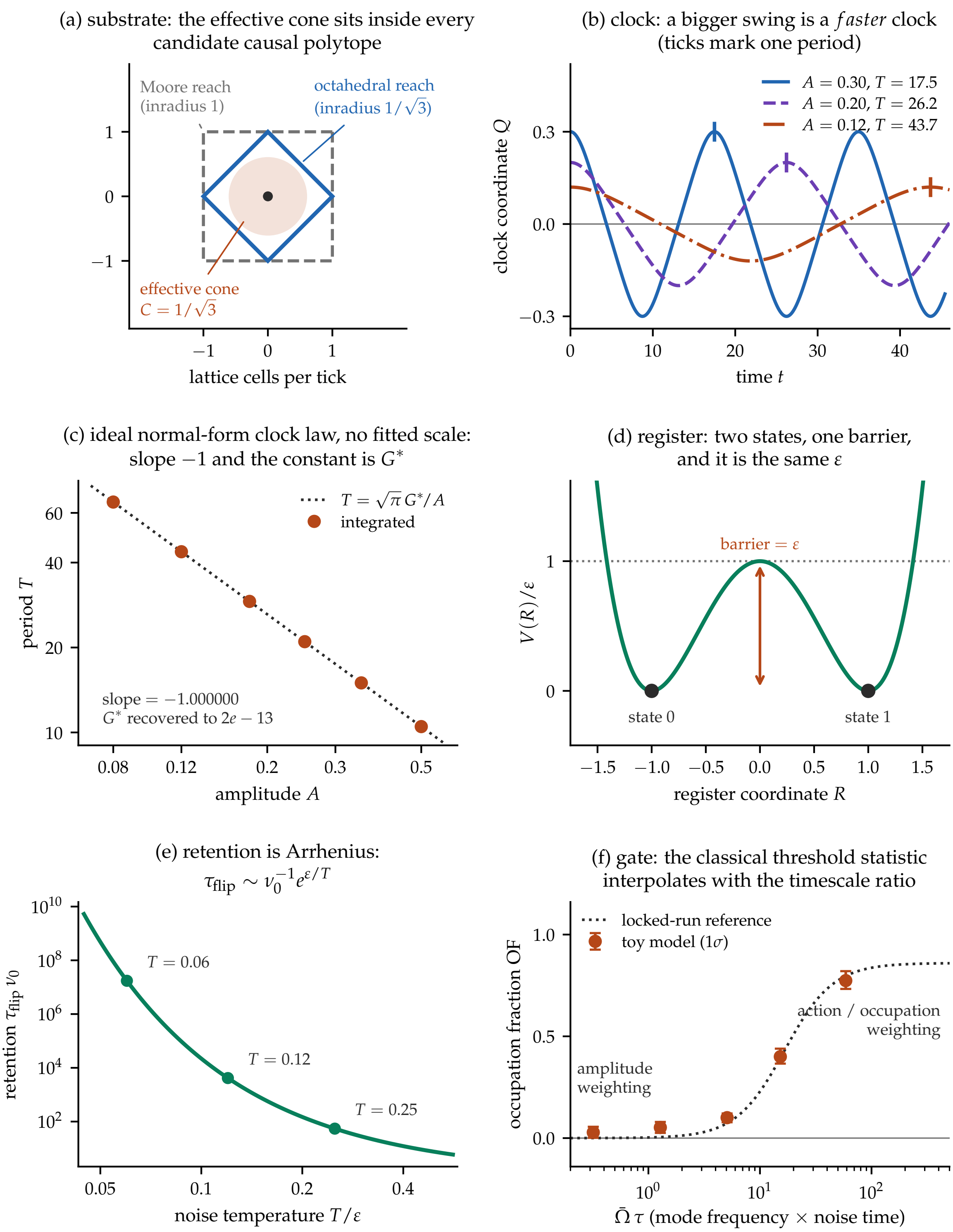

Figure 18: Six minimal temporal-interior modules, computed side by side but not dynamically coupled. (a) The effective cone lies inside the strict reach. (b,c) The ideal quartic normal form has $T \propto A^{-1}$ and recovers G^* to 1.6×10^{-13} without a fitted scale; this does not make the full carrier exactly quartic. (d,e) The compact-support register's barrier equals its well depth and gives the Arrhenius estimate under the stated Kramers assumptions. (f) The classical threshold statistic moves from amplitude toward action/occupation weighting as $\bar{\Omega}\tau$ grows; bars are 1σ bootstrap intervals and the dotted curve is the locked-run descriptive reference of Figure 10.

The demonstrator exhibits two of the three, not all three. In the script, ε is set once and drives the register well and the retention curve — panels (d) and (e), where the barrier is the well depth with no geometric prefactor and retention runs from ~ 55 attempt times at $T = 0.25\,\varepsilon$ to $\sim 1.7 \times 10^7$ at $T = 0.06\,\varepsilon$. The clock panel integrates its own stiffness as an independent hard-coded value, and the detector its own threshold and noise width. So the third role — ε *setting* the clock rate — is asserted by the architecture and *not* exhibited here. Nor could this instrument exhibit it: G^* is recovered from (40) in a form independent of λ, which is what makes the recovery a clean test of the law and a null test of the coupling.

And the economy claim, pressed, does not hold. The clock's stiffness is $\lambda_{\text{eff}} = 8k_1k_2/(k_1 + 3k_2)$. Here k_1 and k_2 are *radial* curvatures. For the registered law (12), $k_1 = 96\varepsilon$. If the purchased closure is declared to have the same shape stretched to radial range $3r_0$, its argument is $q/9$, its depth is an independent ε_2, and $k_2 = \frac{32}{3}\varepsilon_2$. Writing $\rho = \varepsilon_2/\varepsilon$,

$$\lambda_{\text{eff}} = \frac{256\,\varepsilon\,\rho}{3+\rho}, \qquad \Delta E = \varepsilon, \qquad \ln(\tau_{\text{flip}}\nu_0) = \varepsilon/T. \tag{41}$$

Two things follow, and both correct the claim as it was stated. First, the three responses do not move in fixed proportion even at fixed ρ: their exponents in ε are $+1$ for the barrier, $+1$ for the log-retention and $-\frac{1}{2}$ for the clock period, so raising ε deepens the well, lengthens the memory exponentially and *speeds* the clock. Second and more seriously, ρ is a free dimensionless parameter that appears in the clock and in neither of the others: sweeping ρ from $\frac{1}{4}$ to 16 moves $T \cdot A$ by a factor of 3.3 while the barrier and the retention law are untouched. That is precisely the falsifier this subsection wrote for itself — a measurement moving one of the three without the others — and it is fired not by an experiment but by the second interaction species that Section 4.7 had already purchased and priced. **There is a separate clock constant, and its name is ρ.**

What survives is the leading-normal-form statement. In the $A \to 0$ quartic normal form, G^* does not depend on ρ: integrating that reduced mode across $\varepsilon \in [\frac{1}{2}, 2]$ and $\rho \in [\frac{1}{4}, 4]$ recovers $T \cdot A\sqrt{2\lambda_{\text{eff}}/m}/\sqrt{\pi} = G^*$ to 6×10^{-14} throughout. Within that limit, the second energy scale buys the clock's *rate* and cannot touch the normalized quartic coefficient. Finite-amplitude compact-bond motion remains unsimulated, as Section 4.7 states. The economy claim fails where it priced the rate and survives only at leading normal-form order where it priced the number.

11.3 The regime, on an independent instrument

Panel (f) reruns the weighting question with a different construction from the one behind Figure 10. Two standing modes carry lattice-dispersion frequencies $\Omega(k) = 2\arcsin(C\sin(k/2))$ and are given *equal action*, $A_2 = A_1\sqrt{\Omega_1/\Omega_2}$, so they differ in amplitude but not in occupation. The two candidate weightings then predict opposite values for the ratio of second-harmonic coefficients in the excess crossing rate, and the occupation fraction $\text{OF} = (R - R_{\text{amp}})/(1 - R_{\text{amp}})$ is 0 and 1 respectively by construction.

$\bar{\Omega}\tau$	OF (1σ bootstrap)	reference
0.32	0.028 ($-0.026/+0.028$)	0.000
1.28	0.052 ($-0.027/+0.027$)	0.005
5.07	0.100 ($-0.023/+0.021$)	0.073
15.22	0.400 ($-0.034/+0.039$)	0.393
58.77	0.774 ($-0.041/+0.047$)	0.796

The two upper cells agree closely with the locked-run reference; the lower cells sit high by about one standard deviation, which is what a deliberately small instrument looks like when its uncertainty is reported rather than hidden. The qualitative structure — amplitude weighting below the crossover, occupation weighting above it — reproduces on an instrument that shares no code path with the protocol-locked scan.

None of this promotes the detector result into a Born law. The mechanism class is still declared, the ensemble is still imposed, and the engine's own cells still sit at $\bar{\Omega}\tau \in [0.19, 0.86]$, below the crossover, which is why Section 8 reports a null there. What the model shows is narrower and worth stating plainly: the interpolation is not an artefact of one instrument's construction.

11.4 The causal cell in three dimensions

Panel (a) of Figure 18 is drawn in the plane for legibility. Developed properly in three dimensions it yields the paper's sharpest distinction between exact dependency support and a long-wave propagation scale, and it is worth stating on its own.

Two structures compete to define causal reach, and they do not coincide. *Reach* is unconditional: the update rule touches a finite neighbour set each tick, so after t ticks the field is exactly zero outside the t-fold dilation of the causal polytope. This is locality, not dynamics. *The effective cone* is dynamical: long-wave packet peaks propagate at asymptotic speed C. Between the two surfaces sits a shell that is formally reachable but outside that cone. It is useful to track this as a fourth bookkeeping region, but it is not unique to discreteness: dispersive continuum systems also carry precursors outside a group-velocity cone.

The geometry at one tick is exact, and contains one coincidence that is not a coincidence. The octahedron's inradius is $1/\sqrt{3}$, which is the cone speed; the effective-cone sphere is therefore *inscribed* in the octahedron, tangent at its eight faces, and the octahedron is in turn inside the cuboctahedron (inradius 1) and the cube (inradius 1). This is what $C = \min\{\text{containment bounds}\}$ means geometrically: the cone fits inside every candidate causal polytope, and fits the tightest one exactly.

The census is then startling at first reading. **At $t = 1$ the effective cone contains exactly one lattice site — the origin — while eighteen have already been touched.** The nearest site is at distance 1; the cone has radius 0.577. The cone acquires its first other site at $t = 2$, and at $t = 3$ it contains exactly the twenty-seven-site block, because $3C = \sqrt{3}$ is precisely the body-diagonal distance. The correct reading is not that causality fails at one tick, but that the effective-cone sphere contains no non-origin lattice site until the second tick.

What occupies the shell is characterized at the resolution studied by two measurements. First, the group velocity of the leapfrog was differentiated analytically and evaluated on a 181^3 grid of the Brillouin zone. The scan found no value above C and approached C as $k \to 0$; it is evidence for, not a proof of, the global supremum. This does *not* make C the signal front: a supremum over group velocities bounds where the disturbance is *concentrated*, not where it is exactly zero, and the next measurement shows the field is nonzero outside Ct. The strict front is the reach boundary; C is the effective cone. Second, the point-source Green's function at $t = 10$ falls from 5.8×10^{-3} at the cone to 2.8×10^{-13} at the support boundary — ten decades, about 1.3 per lattice cell — and is then identically zero by the finite-support update rule. In that run the shell profile is precursor-like and falls approximately exponentially across the sampled cells. No uniform exponential bound over time and direction is proved. Figure 19 draws the whole structure — the cone inside the reach polytope, the census by tick, and the ten-decade fall to that exact zero.

The cuboctahedral exact support is a structural property of this update; the distinction between support and a group-velocity cone is not unique to discrete systems. In the reported runs the detected contour is much more isotropic than the stencil and its residual anisotropy decreases as a power of elapsed time. Whether that is phenomenologically adequate is not established here.

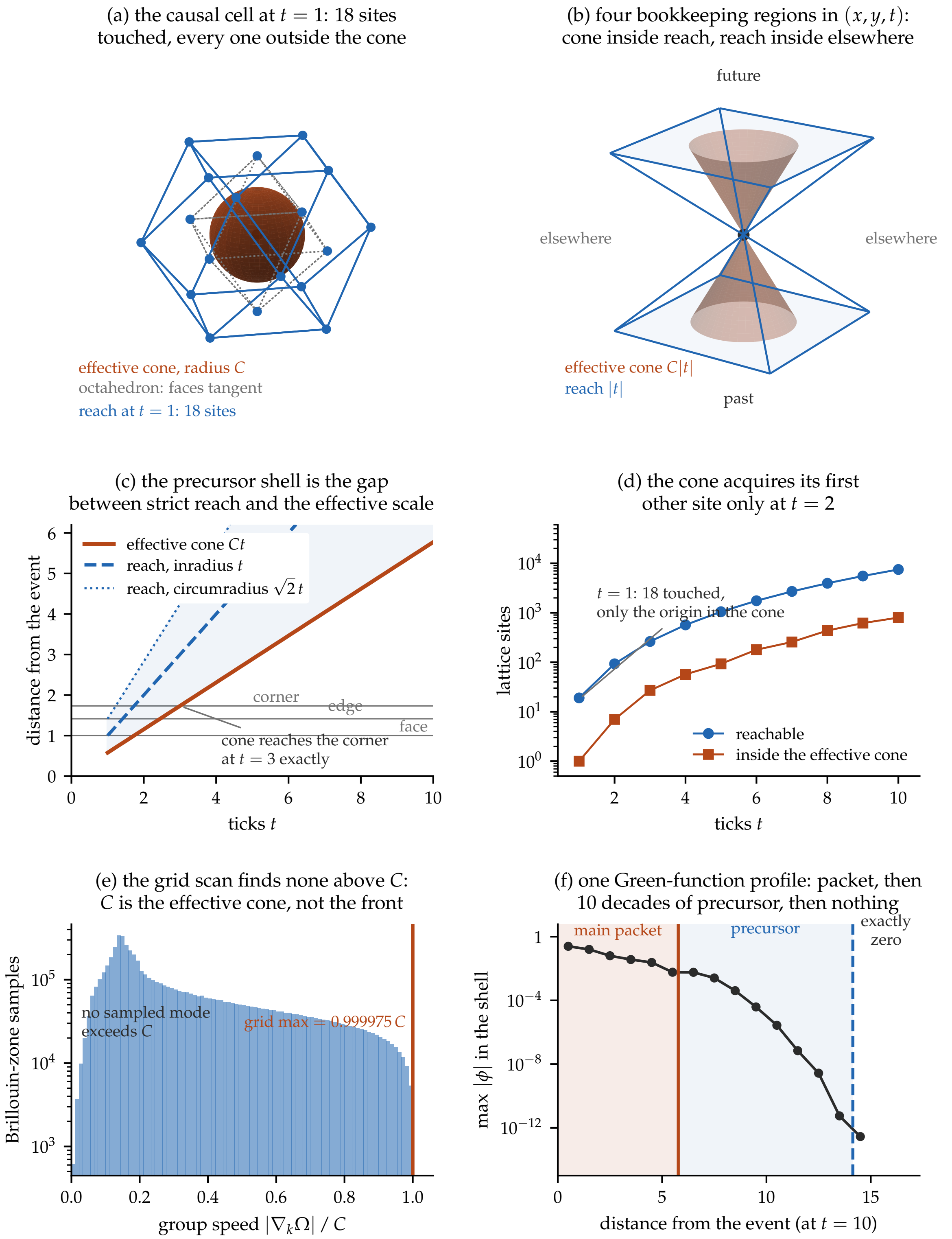

Figure 19: The causal bookkeeping partition around one event. (a) At $t = 1$ the effective-cone sphere is inscribed in the octahedron — its inradius *is* C, so the eight faces are tangent — which is itself inside the reach polytope. (b) Past, future, and elsewhere in (x, y, t), with the round cone strictly inside the square-sectioned reach. (c) The precursor shell is the gap between the two bounds; the cone closes over the body diagonal at $t = 3$ exactly. (d) At $t = 1$ eighteen sites have been touched and only the origin lies in the effective cone. (e) The 181^3 scan finds no sampled group velocity above C; this supports an effective cone but is not a proof of the global supremum. The strict front is the reach boundary. (f) Across the shell the Green's function falls ten decades and then terminates in an exact zero.

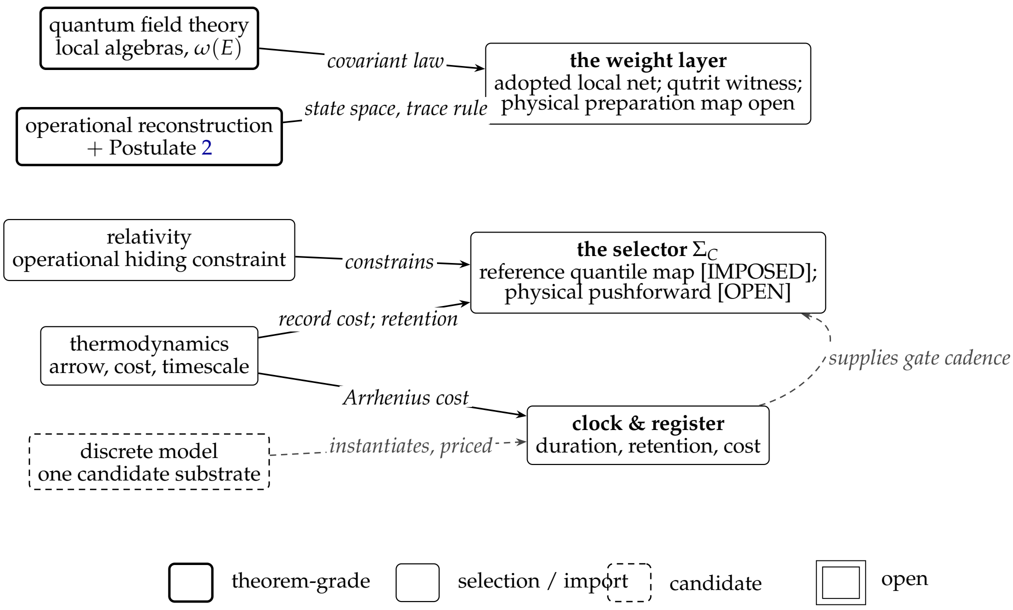

Figure 20: What supplies what. Shapes encode status, never colour, as shown in the legend: solid for theorem-grade within stated premises, thin for selections and imported representations, dashed for a candidate instantiation, and double-ruled for an open interface. Other open interfaces are labelled explicitly in the nodes and caption. Solid arrows are constructions; the dashed arrow marks a candidate instantiation that is priced rather than free. The abstract local net and qutrit witness are adopted at different scopes. Singular actuality and the contextual quantile selector are imposed; the selector proves exact reference pushforward but does not recover physical Born/Bell frequencies.

So the artifact places the calculations in one program and catches a real failure of the advertised constant economy. It does not couple the modules or show them coexisting as one physical system. Nor can it say which pillar supplies which layer, and which layer nobody supplies. That is the statement the next section owes.

12 The merge, stated exactly

We can now say what merges. Figure 20 states it as a supply diagram: which pillar supplies which layer of the architecture, and which layer nobody supplies.

Quantum field theory supplies a covariant comparator for the potential layer [IMPORTED]. Local algebras, covariant evolution, and spacelike order-independence are standard physics. V2 adopts an abstract discrete local net and supplies a finite Type-I qutrit witness. Recovering its preparation map and any physically relevant continuum representation from substrate dynamics remains **open**.

The reconstruction programme plus Postulate 2 supplies trace-form weights conditionally [EXTERNAL THEOREM]. It does not construct a physical substrate measure whose pushforward equals those weights. The Born square is settled inside the imported effect space; the physical Born pushforward remains **open**.

Relativity constrains the selector, and prices each option. The reference Actualization-Batch has commuting local instruments and remote-setting-independent marginals. It does *not* establish empirical hiding of the preferred tick: the gap between (27) and (28) remains a physical recovery debt.

Singular actuality supplies one record [IMPOSED]. Unselected alternatives are potential or counterfactual relative to that record, not additional actual histories. Identifying the model's manifestation rule with the selector is a priced conjecture, not a theorem.

Thermodynamics supplies the record bill, not the statistics. The implemented update is operationally lossy and the register has a priced retention time. A complete entropy and reset account is open. None of these facts determines the relative frequencies of alternative records.

Bell supplies an independent cross-context obstruction [THEOREM]. One common substrate space carrying four jointly defined local response variables under one setting-independent ensemble measure implies $|S| \leq 2$, however singular, dissipative, or noninvertible each record is. The imposed reference selector explicitly rejects joint local response factorization while preserving measurement independence; it reaches the selected quantum correlations only because it consumes their weights.

The discrete model instantiates the architecture and prices it. It supplies succession, a strict dependency reach, screened harmonic clock candidates, and candidate manifestation events. It currently realizes the leading-order lemniscatic clock only in the purchased construction; no native quartic clock has been found. Retention likewise requires the declared register and bath assumptions. The slow-channel experiment supplies only a classical detector regime.

The merge is therefore an itemized architecture, not a unification: quantum theory specifies conditional weights, thermodynamics prices the production and retention of a record, relativity constrains its operational description, and the reference selector shows that deterministic contextual actualization is logically compatible with the Bell accounting. The central open object is the joint *physical* construction of the preparation map, equilibrium measure, response map, and required Born/Bell pushforwards without reading their targets.

Itemized is a claim that can be checked, and checking it needs the items in one place at their actual strength rather than at the strength the surrounding prose lends them. That table is next, and it is the only part of this paper that binds.

13 Status of claims

Table 2 states every substantive claim in the paper at its actual strength. Where the body and this table disagree, the table is correct.

Table 2: Every substantive claim in the paper, at its actual strength.

Claim	Status
The actual/potential distinction (2)	**[AXIOM — semantic primitive]**; organizes the scaffold but supplies no dynamics or frequencies
Singular actuality	**[IMPOSED — Postulate 1]**; one mutually exclusive outcome enters the record; not derived from the model
Unselected alternatives remain potential/counterfactual	True by definition within the scaffold; no theorem that they persist as independent substrate objects
Closure map Γ from history to state	Declared scaffold; history alone is insufficient
Actual record algebra (14)	**[AXIOM]**; bounded classical configuration functions are commutative and finite ternary records form $\mathbb{C}^{3^{|\Lambda|}}$, not a full matrix algebra
Potentiality net and qutrit witness (15)	**[AXIOM] + [SELECTED REFERENCE REPRESENTATION]**; finite regions are Type I and the quasi-local reference limit is UHF. No automatic Type-III claim
Preparation map Γ_Λ	Adopted interface with restriction consistency proved in the reference witness; physical substrate recovery **[OPEN]**

Table 2, continued

Claim	**Status**
Identification of genesis/manifestation with Σ_O	**[CONJECTURE — model-scoped identification]**; the engine rule is explicit, but no laboratory outcome map is constructed
Genesis readout is many-to-one	**[MEASURED — engine-scoped]**; distinct flux magnitudes yield the same discrete record while the continuous flux remains stored
Complete update-map noninjectivity	**[THEOREM — current-engine scope]**; witnessed by evaporation, not by genesis alone and not for every event
Resource price of the candidate record	**[MEASURED] + [THEOREM]**; in the declared model, accepted genesis has a positive drain, and a reusable symplectic energy lift requires added record-bearing bath variables plus reset, replacement, export, or feedback. No universal energy quantum or complete Landauer cycle is derived
Operational framework and six principles	Imported premises; not derived here
Complex finite-dimensional quantum theory	External theorem, conditional on those premises [1]
Convex state/effect structure	Derived within the imported framework
Effect-noncontextual additivity	Declared postulate (Post. 2)
Trace rule and the Born square (20)	External **[THEOREM]**, conditional on the imported effect space and Postulate 2 [2]
Discrete local-net proof package	**[THEOREM — adopted reference net]**; isotony, actual commutativity, spacelike commutation, compatible state restriction, normalized causal instruments, and order independence
Physical Born pushforward (22)	**[OPEN]**; neither the physical response map nor its ensemble measure is constructed
Deterministic quantile selector (23)	**[IMPOSED REFERENCE REALIZATION]**; exact pushforward under invariant Lebesgue measure, measurement independent and globally contextual. It consumes $\omega(E)$ and is therefore not a physical Born derivation
CPTP dynamics, unitarity, Schrödinger form	External theorems, conditional on their stated premises
Quadrature root $\times$ degree ratio (4)	Exact identity within the homogeneous family $V = \lambda|x|^n$; it is not a universal factorization for arbitrary mechanical potentials. The harmonic case is uniquely self-reproducing within that family
Threshold forcing (Prop. 1)	Exact within H1–H4 and only as $A \to 0$. It neither places a substrate at the codimension-one threshold nor proves finite-amplitude quarticity
Finite-amplitude clock drift	The sextic family (7) is exact for that family. An exploratory harmonic-spring surrogate gives $+0.33\%$: its reduced quadrature assigns $+0.314\%$ to amplitude-dependent mass and $+0.014\%$ to potential at $A = 0.12$. No compact-law finite-amplitude result is supplied

Table 2, continued

Claim	Status
G^* **is chosen, not derived**	No valid relativistic carrier has produced it. The native screen was negative; the purchased four-body carrier is only asymptotically quartic and Galilean; the soliton shape-mode evidence is unreproduced and isochronous
The rigidity trichotomy and buckling criterion	Exact within their declared mechanical model and hypotheses
Minimum viable leading-order quartic carrier	Minimal four-body/two-scale construction and $A \to 0$ normal form exact. Finite-amplitude numbers come only from a curvature-matched harmonic surrogate; G^* recovered to 2×10^{-6} there only by extrapolation. Galilean and disqualified as a relativity test
No native quartic clock	Negative at screened scope only; not a no-go theorem
Context-blind dual clock feedback and compliant gate	**[SELECTED REFERENCE CONTROLLER]**; detuning and amplitude/action channels with explicit work, dissipation, and compliance audit. Native actuation and non-rescalability remain **[OPEN]**
Register barrier = ε (Prop. 2)	**[THEOREM — declared compact-support model]**; exact plane crossing and $N \leq 2$ bound plus 11,250,000 exact-rational, outward-rounded boxes, zero samples and zero uncertified. Conditional on compact support; a Morse/LJ tail lowers the barrier
One constant, three roles	**Refuted as stated.** $\rho = \varepsilon_2/\varepsilon$ moves the clock rate alone; in the leading $A \to 0$ quartic normal form, G^* is independent of ρ to 6×10^{-14}. No finite-amplitude compact-law result
Threshold interpolation toward action/occupation weighting	**[MEASURED — mechanism-scoped, imposed ensemble]**; OF is a regression coordinate, not a probability. Neither an exact Born law nor a substrate theorem
Non-transfer to the model's thermal field	Protocol-locked negative at sampled scope; same-band diagnosis post-hoc
Purity requirement	Measured on five shares/32 seeds in one declared additive instrument; not a universal proposition
Slow sector is slow ($\tau = 317.9$ vs 0.81)	Measured profile; deterministic, so seed-independence is vacuous
Common-measure CHSH bound (Prop. 3)	**[THEOREM]**; four jointly defined responses under one setting-independent measure imply $\|S\| \leq 2$. Record invertibility and energetic cost are not premises
Singularity, dissipation, or noninvertibility *alone* as an escape from Fine/Bell	**[CLOSED NEGATIVE]**; it changes record formation, not the induced joint distribution
Reference CHSH recovery by Σ_C	**[THEOREM inside adopted algebra + IMPOSED selector]**; selected singlet/observables give $2\sqrt{2}$, all local marginals are no-signalling, local factorization fails, and PR-box $S = 4$ is excluded. No physical substrate Born/Bell recovery

Table 2, continued

Claim	Status
Order-independence (27) $\not\Rightarrow$ equivariance (28)	Logical separation; order independence passes in the adopted causal-instrument reference model, while empirical preferred-order hiding and strong history equivariance remain open
Operational hiding of a preferred foliation	Conditional reduction (Prop. 4); follows if quantum composed-history statistics and spacelike order-independence are already supplied
Free scalar spatial stencil	**[THEOREM — tree level]**; unique face/edge weights yield normalized isotropy through $O(k^4)$, with leading directional residual $(ka)^4/3240$
Radiative stability / dimension-four protection	**[OPEN — HARD]**; cubic symmetry does not relate temporal/spatial, electric/magnetic, native-vector, or inter-sector coefficients. One frozen QED-like branch requires a dimension-four counterterm; no on-shell closure is shown
Spatial rotational isotropy	Not closed generally. Only the declared scalar spatial symbol is improved at tree level; the complete interacting operator inventory remains open
Strict causal support of the M18 wave update	**[DERIVED]**; face and edge sites are nonzero at $t = 1$, corners zero. The support boundary is the M18 reach polytope, not Ct
Effective Green-function contours	**[MEASURED]**; near-Ct threshold contour has 0.71% anisotropy and sampled scaling $t^{-1.1}$–$t^{-1.3}$. Not the exact causal surface or a uniform theorem
Which neighbourhood defines causal reach	A model choice with exactly distinguishable consequences; M6, M18 and M26 differ at the first tick. Finite-run precursor suppression does not erase exact support
Free massive scalar modes	**[DERIVED — toy sector]**; numerically checked; $C_{\text{eff}}/C = 1 + M^2/12 + \cdots$ and residual $M^2\beta^4/4$. Electron/proton numbers are parametric evaluations, not physical predictions
Free cross-sector symbol match	Exact integer-hop eighteen-square and spatially cubic-covariant half-angle seven-square constructions. Half offsets are therefore an economy, not an existence requirement. The latter gives $n_{\text{cov}} \leq 7$ but not a proven cubic-covariant minimum; mixed-sector five/six remain unexcluded. A non-covariant four-square numerical candidate has residual 5.3×10^{-15} but no exact/interval certificate, so the rigorous bounds are $3 \leq n_{\text{min}} \leq 7$. A working matter sector, time discretization and interactions are not built; current production sectors still mismatch. If that economical half-offset construction is adopted, its ontology remains **[OPEN]**: refined physical sites recalibrate a_{phys}, while staggered cochains keep a but add carrier types; neither reading is adopted

Table 2, continued

Claim	Status
Free multiparticle-threshold inheritance	**[DERIVED — FREE THRESHOLD]**; (36) and (37) checked to reported numerical precision. Not a bound-state or macroscopic theorem; nucleon/Earth extrapolation withdrawn
Static-well composite clock	**[MEASURED — exploratory, inconclusive]**; the old $p = -1$ target used an infinitesimal-gap approximation. Against the exact finite-gap comparator, six fixed cases at $K = 0.2$ differ by 0–1.1%; convergence and held-out-momentum controls remain open
φ^4 soliton shape-mode clock	**[OPEN — UNREPRODUCED; COMMON-CONE MISMATCH]**; $p \simeq -1$ requires a fitted speed about 6% below the energy cone, and the committed runner does not reproduce the quoted campaign
Interacting vertices, bound-state cones, composite boosts	**[OPEN]**; requires one interacting action, stress-energy, and boost generator
Gravitational actualization resemblance	Speculative alignment only; the parameter-free version is externally excluded [35]
$\hbar$, Hamiltonian, masses, couplings	**Not derived** anywhere in this paper

14 Open problems

1. **The Born pushforward for a concrete substrate.** Construct F_O and μ_{eq} from the substrate's own dynamics without reading target probabilities. The locked, unexecuted first campaign tests the free-flux density $n = |\phi_+|^2$, equilibrium invariance, context overlap, held-out instruments, and an explicit target-leak sentinel; interacting and genesis transfer are later gates.

2. **The physical Bell pushforward.** V2 has already named the rejected premise: the deterministic selector is globally contextual and nonfactorizable, while measurement independence is retained. Recover that response law from the substrate and show no-signalling on held-out contexts. Noninvertibility, dissipation, and singular records do not answer this.

3. **Thermodynamic closure of record writing.** Complete a repeated genesis–record–reset cycle with bath state, heat, work and entropy tracked. The current bath-rank and non-repeatability theorems price the missing machinery but do not supply an autonomous cycle or a universal cost per outcome.

4. **Native quartic timekeeping.** Does a finite integer unit-strut tensegrity exist under the single-scale law, with blocking form definite on its full flex space? No candidate realizes it; no impossibility proof exists.

5. **What physically holds the threshold.** Proposition 1 forces the quartic *at* $\mu = 0$, and $\mu = 0$ is a codimension-one surface: the generic configuration is not on it, and an unmaintained system falls off it and becomes a harmonic clock. The selected dual feedback controller proves that a context-blind audited loop can hold the reference clock. What native mechanism pays that work, survives held-out perturbations, and distinguishes G^* from a removable period rescaling?

6. **Shielded actualization.** The measured purity diagnostic (page 23) indicates that, in the tested additive instrument, the slow channel must be nearly exclusive at the threshold. Can any physical manifestation rule be shielded from its own field's fast noise?

7. **Preferred-order operational hiding.** Strong history equivariance is not imposed by v2; its ontology permits a real preferred order. Operational hiding is conditionally reduced by Proposition 4, but the substrate Born/Bell hypothesis and common-cone tests remain open.

8. **Quantum sequential growth.** The classical classification is known; the quantum analogue with interfering weights is open.

9. **A physically coupled model.** The v2 standalone simulator now composes history, preparation, local instruments, maintained G^* gate, selector, and records, but it imports the state–effect weights. Recover the preparation and pushforward from substrate dynamics, add register-to-carrier backreaction, and check the energy and signalling audits without target leakage.

10. **What the second energy scale is.** $\rho = \varepsilon_2/\varepsilon$ is now a named free parameter of the architecture (41) rather than an unnoticed one. Either derive it from the closure species' own structure, or accept it as a second declared constant and price it in the import ledger.

11. **The barrier without compact support.** Proposition 2 is exact because the bond law vanishes identically beyond 3/2. Establish the corresponding statement for a potential with a tail — where the barrier is strictly below ε — or show that compact support is forced.

12. **From the selected net to a physical representation.** The finite qutrit witness is Type I and its quasi-local limit is UHF. Determine which preparation states the substrate generates, construct the relevant GNS representations, and classify any limiting factor only from that state-and-representation data. No Type-III status is assumed.

13. **Radiative and interacting common-cone matching.** Inventory all symmetry-allowed operators of dimensions three and four for one action, remove the single time-normalization redundancy, and compute the remaining on-shell relative-cone eigenvalues. A required counterterm is imposed, not emergent.

14. **Bound-state and soliton shape-mode recovery.** Compute $E_B(K)$ for a real interacting bound state rather than the free threshold (36). Separately restore a locked, reproducing soliton shape-mode runner and test a comoving internal-mode projection, radiation and finite-volume convergence, continuum scaling, and one independently fixed shape-mode/energy cone on held-out velocities.

A Instruments, protocol records, and provenance

Every measurement *in the series catalogued below* has an author-attested protocol record stating that outcomes, gates, kill conditions, and runner hashes were fixed before execution. The current hashes match those records, but the ordering is not corroborated by an immutable external timestamp, as the next paragraph explains. The protocol discipline still caught one invalid instrument: a mis-derived conservation invariant failed at 3.6×10^{-2} against a 10^{-10} tolerance, and the instrument was re-locked and re-run before producing a reported number.

What that discipline does and does not currently establish. Stated plainly, because the distinction is easy to overclaim. The recorded hashes verify that the runner in the repository *now* is byte-identical to the one the record names, and they do. What they do not establish is the *ordering*: the records and their runners entered version control in commits made *after* the runs they govern, and the git tag that would have locked them beforehand — a discipline this project uses elsewhere, with 136 such tags — was never minted for this arc. So pre-registration here is attested by the records' own dates and an interim hash, not corroborated by an immutable external timestamp. A reader who declines to take the dates on trust should treat these as carefully documented analyses rather than as externally audited pre-registrations. Separately and more seriously: the committed soliton shape-mode runner does *not* reproduce the parameter set quoted for the shape-mode dilation campaign in Section 9.5.6 — it carries a superseded first-pass configuration whose wider couplings lose the shape mode entirely, and re-running it does not return the exponents printed there. Those exponents should be read as unreproduced from the repository as it stands until that runner is restored.

Exact protocol-to-runner mappings. The following are repository-relative paths; each SHA-256 is the current runner hash that matches its named record. These mappings establish present byte identity, subject to the ordering caveat above.

Weighting discrimination. Record: `docs/theory/10_eft_program/preregistrations/quantum_foundations/PREREG_BORN_DENSITY_UPCROSSING_v1.md`.
Runner: `scripts/experiments/verify_born_density_upcrossing.py`.
SHA-256: `269E3A2DA015A5515FA25C966BEC573ACF9F26408F8D7E0504BFB4CE8C60F690`.

Saturation scan. Record: `docs/theory/10_eft_program/preregistrations/quantum_foundations/PREREG_BORN_DENSITY_SATURATION_v2.md`.
Runner: `scripts/experiments/verify_born_saturation_v2.py`.
SHA-256: `2887218838B7B6D9A49F8EFF04617A8210CE7BB9B4096ED1F43E92E0DB283ED4`.

Instrument	Design	Outcome
Weighting discrimination	two modes, equal occupation, 48 seeds	Regime-dependent: OF 0.024 $\to$ 0.099
Saturation scan	five arms, $\bar{\Omega}\tau = 0.64$–78	Approaches occupation endpoint: OF $\to$ 0.836, monotone
Engine regime map	ten cells, native thermal field	**No structure**: $\|\rho\| \leq 0.37 < 0.7$
Slow-sector timescale	self-gravitating configuration	Slow: $\tau = 317.9$ vs 0.81, ratio 391
Slow-channel coupling	five slow-shares, 32 seeds	Instrument result: OF = 0.936 at $f = 1$

Table 3: The Section 8 instrument series. Full parameters, seeds and gate values are in the corresponding protocol records; the exact record–runner–hash mappings are printed below. The composed model of Section 11 is deliberately *not* listed: it is a demonstrator that reproduces known answers, not an instrument of record, and no claim in this paper rests on it.

Engine regime map. Record: `docs/theory/10_eft_program/preregistrations/quantum_foundations/PREREG_BORN_REGIME_MAP_ENGINE_v1.md`.
Runner: `engine/tests/campaign_born_regime_map.cpp`.
SHA-256: `90DE28E7459458171154A41CD277A6E3F37586658A5A3063F4F3447438CD2EA7`.

Slow-sector timescale. Record: `docs/theory/10_eft_program/preregistrations/quantum_foundations/PREREG_LATENCY_SLOW_GATE_v1.md`.
Runner: `engine/tests/campaign_latency_slow_gate.cpp`.
SHA-256: `1F7B6AAC9D7F8C7BA935869B8A240F6DD51B2777BF036144E9D73FFB96EC6D82`.

Slow-channel coupling. Record: `docs/theory/10_eft_program/preregistrations/quantum_foundations/PREREG_LATENCY_SLOW_GATE_v1.md`.
Runner: `scripts/experiments/temporal_interior/verify_latency_gate_mechanism.py`.
SHA-256: `1E58D24D1F92F88E197FD16C779DE1F5D1462BAE9C4E8228D9FDF8F822B59F0D`.

Analytic artifacts, which are proofs rather than instruments. Three results in this paper are established by argument and verified by computation rather than measured, so they carry no pre-registration and belong outside the series above. All three run in seconds and print their own assertions. Proposition 2, both bounds, is `scripts/experiments/temporal_interior/derive_register_barrier_lower_bound.py`; the economy result (41), including the exponents and the ρ-independence of G^* to 6×10^{-14} in the leading $A \to 0$ quartic normal form, is `derive_epsilon_economy.py` in the same directory. The period decomposition of Section 4.2 is `derive_period_decomposition.py`.

Exploratory clock provenance — not pre-registered. The finite-amplitude harmonic-spring surrogate values in Section 4 come from the following artifacts:
Analysis: `docs/theory/10_eft_program/native_time_carrier_programme/ANALYSIS_MINIMUM_VIABLE_CLOCK_CARRIER_v1.md`.
Runner: `scripts/experiments/mvc_fourchain_clock.py`.
Current runner SHA-256:
`E2AEAE0B73D311113F7944D053AD8AD1AD5E0AF2567687F9F7B321921E80F789`.

This is explicitly exploratory and **not pre-registered**. `dissemination/papers/semantic_ontology/figures/make_figures.py` transcribes the reported 2×10^{-6} and 3.3×10^{-3} plotting values; it does not recompute that surrogate campaign.
The clock's falsifier was stated before any engine campaign. A seeded four-chain must show the anti-pendulum signature, meaning unit slope of $\log f$ against $\log A$, and must satisfy

$$TA\sqrt{2\lambda_{\text{eff}}/m_{\text{eff}}}/\sqrt{\pi} = G^*$$

to declared tolerance, in a profile where no imposed phenomenology does the work. Failure of either kills the candidate.

B The reconstruction as an import invoice

Treating "the Hilbert space" as one unpriced assumption hides its structure. The reconstruction replaces it with six separately auditable premises, and therefore with six separately falsifiable questions that can be put to a candidate substrate one at a time. The scoring below is preliminary and non-binding; it names what would have to be shown.

Premise	What a substrate would have to establish
Causality	Plausibly native given forward determinism and an arrow; the record-level protocol test is distinct and open
Perfect distinguishability	Plausibly native for discrete states; whether this carries the reconstructed-state meaning is open
Ideal compression	Open; no analogue scored
Local distinguishability	**The complex-selecting premise** — real-Hilbert theory fails it. A native complex structure in the lattice symmetry is suggestive but does not establish operational local tomography
Pure conditioning	Open; no analogue scored
Purification	**The quantum-selecting premise, and the likely crux import** — whether a dispositional layer can serve as purifier structure is a genuine research question

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